

Copy control and other control properties in Mandarin

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Abstract

The possibility of overt controllees has long challenged control theories. This paper investigates copy control in Mandarin Chinese, where the embedded subject of a control construction may be overtly realized. Based on acceptability-judgment experiments, we establish empirical generalizations concerning how copy control interacts with exhaustive control, partial control, and raising. We show that copy control is available in Mandarin, but only under tightly constrained conditions: it is licensed with exhaustive and partial control verbs in specific configurations, but excluded with raising predicates and transitive subject-control verbs. We argue that these patterns are not best captured by finiteness-based approaches, which dominate existing accounts. Instead, we propose a formal analysis within Lexical Functional Grammar (LFG), distinguishing functional control (structure sharing) from anaphoric control (pronominal dependency). Copy control arises precisely in anaphoric-control environments, while being systematically excluded under functional control. The paper contributes to cross-framework comparison (LFG vs Minimalist approaches), distinctions between grammatically licensed copy control and processing-based resumptives, lexical specification under interspeaker variation, and the typology of anaphoric control. We argue that a hybrid model integrating structure sharing and anaphoric dependency provides a principled and empirically grounded account of complex control phenomena.

Keywords. Control theory; copy control; Mandarin Chinese; Lexical Functional Grammar; experimental syntax

1 Introduction

For decades, control theory has played a central role in linguistic theorizing, at least since Rosenbaum (1967). Within the Principles-and-Parameters and Minimalist research programs, two major approaches to control are widely recognized: the *PRO-based theory* (see, e.g., Chomsky 1981; Landau 2004, 2013, 2015) and the *movement theory of control* (see, e.g., Boeckx et al. 2010; Hornstein 1999, 2001; Hornstein & Polinsky 2010).¹ The PRO-based theory and the movement theory of control are often regarded as competing theories with substantial differences in theoretical mechanisms and empirical coverage (Potsdam & Haddad 2017). On the other hand, Lexical Functional Grammar (LFG) has posited two major

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¹In the Minimalist era, the PRO-based theory is typically associated with Landau's work, including his Agree-based theory of PRO and, more recently, his two-tiered theory (see Section 5.4). Another more recent approach is the form-copy theory of control (Chomsky et al. 2023), which shares important similarities with Rosenbaum's (1967) Equi-NP deletion and the movement theory of control. See Landau (2024b) for a critique of the form-copy theory.

approaches to analyzing control phenomena since Bresnan (1982a): *functional control* and *anaphoric control*. Functional control employs syntactic sharing to account for the interpretive dependency between the controller and the controllee, whereas anaphoric control treats the controllee as a pronominal, subject to binding constraints. For the purpose of cross-framework comparison, one may draw parallels between anaphoric control and the PRO-based theory of control on one hand, and between functional control and the movement theory of control on the other hand, with the caveat that LFG is a non-movement theory (Vincent 2023: 610–611). Rather than being competing approaches, LFG’s functional and anaphoric control often co-exist in a given language and thus complement each other in capturing the full complexity of control phenomena in the language (Dalrymple et al. 2019: ch. 15).

In research on control in Mandarin Chinese, there has been an ongoing debate surrounding an intriguing phenomenon known as copy control, which—pre-theoretically—refers to an empirical pattern in which the controllee appears overtly, contrary to its canonical unexpressed form (see, e.g., Grano 2015; Hu et al. 2001; Li 2024; Zhang 2016). Example (1) is an instance of Mandarin copy control, with an object-control verb *jiao* ‘ask’, where the controllee (underlined in the example) appears as an optional overt personal pronoun coreferring with the controller *Zhangsan*. As noted by Grano (2015); Polinsky & Potsdam (2006); Zhang (2016), among others, the form of the controlled copy varies cross-linguistically: it can appear as an overt pronoun or full noun phrase, and can be obligatorily or optionally overt. In Zapotec (2) the controllee is an optional full noun phrase; in Ewe (3), it is an overt logophoric pronoun.² In Hungarian (4) it can be a complemented pronoun (comprising a person pronoun and a noun complement).³ While examples (1)–(4) illustrate copy control in a complement clause, Telugu (5) allows copy control in an adjunct-control structure.^{4,5}

- (1) Mandarin Chinese (Hu et al. 2001: 1132)

wo_i jiao zhangsan_j jintian xiauwu (ta_{*i/j/*k}) wulunruhe dou yao lai
 I ask Zhangsan today afternoon 3SG in.any.case all will visit
 ‘I asked Zhangsan to visit this afternoon, no matter what happens.’

- (2) San Lucas Quiavini Zapotec (Lee 2003: 102)

R-càà’z Gye’eihly_i g-auh (Gye’eihly_i) bxaady
 HAB-want Mike IRR-eat Mike grasshopper
 ‘Mike wants to eat a grasshopper.’

- (3) Ewe (Satik 2021: 10)

Agbe_i dzagbagba/ ɲlobe/ dzina/ vɔvɔm/ wosusu/ dzi/ susum be yè_i-a kpo
 Agbe try/ forget/ want/ afraid/ decide/ like/ intend C LOG-IRR experience
 dzidzɔ
 happiness
 ‘Kofi_i tried/forgot/wanted/is afraid/decided/likes/intends PRO_i to be happy.’

²Satik (2021) analyzes the logophoric pronoun as an overt PRO.

³We have added brackets to Szabolcsi’s (2009b) original data to show that the complemented pronoun is optional. We thank György Rákosi and Péter Szűcs for discussion of this example.

⁴Since Telugu allows both forward and backward control, it is debatable whether it is the matrix subject or the subject in the adjunct clause that serves as the controllee. Thus, both subjects are underlined. According to Haddad (2009: 84), preserving either subject is sufficient; although some Telugu speakers consider pronouncing both subjects as redundant, they do not regard the copy-control structure as unacceptable.

⁵Abbreviations: 1 = 1st person; 2 = 2nd person; 3 = 3rd person; ACC = accusative; ADV = adverbial; BA: object-fronting device; C = complementizer; CNP = conjunctive participle; DAT = dative; DE = pre-nominal modification marker; DUR = durative; EC = exhaustive control; M = masculine; NOM = nominative; PART = participle; PC = partial control; PRF = perfect; PL = plural; HAB = habitual; INF = infinitive; IRR = irrealis; LOG = logophor; SG = singular.

- (4) Hungarian (Szabolcsi 2009b: Ex. (49))
 Szeretné-nk (csak mi nyelvészek) kap-ni magasabb fizetés-t
 would.like-1PL (only we linguists) get-INF higher salary-ACC
 ‘We would like it to be the case that only we linguists get a higher salary.’
- (5) Telugu (Haddad 2009: 70)
 [(Kumaar-ki_i) aakali wees-i] (atanu_i/ aa pichooDu_i/ Kumaar_i) saandwic tinnaa-Du
 Kumar-DAT hunger.NOM fall-CNP (he/ that idiot/ Kumar.NOM) sandwich ate-3.M.SG
 ‘Having got hungry, he/that idiot/Kumar ate a sandwich.’

Set against the backdrop of a robust cross-linguistic empirical landscape of copy control, the discussion of copy control in Mandarin has generated considerable controversy over the years, spanning from basic questions regarding the acceptability of the structure to its broader theoretical implications. As a case in point, two recently published papers, namely Huang (2024) and Li (2024), cannot even agree upon the basic empirical data of Mandarin copy control, not to mention deeper theoretical issues. Moreover, much discussion to date relates Mandarin copy control to unresolved claims about the nature of finiteness in Chinese (see, e.g., Hu et al. 2001; Huang 1989; Li 2024; Xu 2003; Zhang 2016). Thus, there is still much room for investigation of copy control in Chinese, including both its empirical foundation and its theoretical analysis.

This paper aims to illuminate the properties of copy control in Mandarin Chinese and its interactions with other control-related properties in the language. We argue that copy control plays a demarcating role in the grammar of Mandarin by revealing two distinct types of control machinery: one involving syntactic structure sharing, and the other, anaphoric dependency. Contrary to previous studies relating Mandarin copy control to finiteness principles, we propose that LFG’s functional vs. anaphoric control distinction provides a more compelling explanation (see, e.g., Dalrymple et al. 2018, 2019; Haug 2013, 2014). In support of this view, we present new empirical generalizations informed by new acceptability-judgment experiments.

The paper is organized as follows. Section 2 reviews past debates on Mandarin copy control. Section 3 discusses the empirical patterns and the acceptability-judgment experiments that help establish these patterns. Section 4 offers a formal analysis of Mandarin copy control. Section 5 discusses the broader theoretical implications of our analysis, and Section 6 concludes the paper.

2 Previous work on overt controllees in Mandarin Chinese

Mandarin is a discourse pro-drop language (Huang 1984, 1989; Roberts & Holmberg 2010). (6a) is a non-control structure containing an unexpressed subject in the complement clause (marked as “ $\emptyset$ ”), which can refer to any discourse-salient entity. (6b) is a control structure where the unexpressed subject is restricted to corefer with the matrix subject. Our paper focuses on control structures.

- (6) a. Xiaoming_i shuo $\emptyset_{i/j}$ yao kan shu
 Xiaoming say $\emptyset$ will read book
 ‘Xiaoming_i said (he/she/they)_{i/j} will read books.’ (non-control structure)
- b. Xiaoming_i shefa $\emptyset_{i/*j}$ zuo hao ren
 Xiaoming try $\emptyset$ be good person
 ‘Xiaoming tries to be a good person.’ (control structure)

2.1 Finiteness

Early discussion of the overtness of embedded subjects in Mandarin is provided by Huang (1982, 1984, 1989) and Li (1985, 1990). Since then, most, if not all, of the discussion on this topic has centered around the question of finiteness in Mandarin, which lacks verbal inflection for tense and person–number agreement. The contrast between (7a) and (7b) and between (8a) and (8b) has been attributed to an abstract finiteness distinction, where the controlled complements in (7a) and (8a) are non-finite and thus cannot host overt subjects, whereas the non-controlled complements in (7b) and (8b) are finite and therefore can contain overt subjects.

- (7) a. wo zhunbei (*wo) mingtian lai
I prepare I tomorrow visit
'I prepare to visit tomorrow.' (Huang 1982: 249)
- b. wo shuo (wo) mingtian lai
I say I tomorrow visit
'I said I will visit tomorrow.'
- (8) a. wo quan zhangsan (*ta) bu mai zhe-ben shu
I persuade Zhangsan 3SG not buy this-CL book
'I persuaded Zhangsan not to buy this book.' (Huang 1982: 249)
- b. wo gaosu zhangsan_i (ta_{i/j}) bu-yao mai zhe-ben shu
I tell Zhangsan 3SG not-will buy this-CL book
'I told Zhangsan that he wouldn't buy this book.'

2.2 Intervening adverbial phrase

However, Hu et al. (2001) demonstrate that it is in fact possible for (7a) and (8a) to contain overt embedded subjects, provided that there is an intervening adverbial between the matrix verb and embedded verb, as in (9a) and (9b).⁶

- (9) a. wo zhunbei [mingtian xiawu tian hei yihou] (wo) yi-ge ren lai
I prepare tomorrow afternoon sky dark after I one-CL person visit
'I prepare to visit on my own tomorrow after it gets dark.' (Hu et al. 2001: 1131)
- b. wo quan zhangsan_i [ruguo mei you ren mai zhe-ben shu] (ta_i) ye mai yao
I persuade Zhangsan if no have people buy this-CL book 3SG also not will
mai
buy
'I persuaded Zhangsan not to buy this book if no one bought it.' (Hu et al. 2001: 1131)

Hu et al. (2001) argue that the unacceptability of the overt pronouns in (7a) and (8a) is not linked to finiteness, but is the result of an obviation principle which states that an overt pronoun tends to obviate from its closest prominent noun phrase (Hu et al. 2001: 1134). In (9a) and (9b), there is an intervening adverbial between the overt pronominal subject and the controller, and so the controller is no longer the closest prominent noun phrase to the pronoun, thereby satisfying the obviation principle

⁶In English, *persuade* has an alternative non-control complementation structure, as in *John persuaded Sarah that Fred was wrong*, but Mandarin does not have this kind of non-control version of *quan* 'persuade'.

and accounting for the acceptability of (9a) and (9b).⁷ Hu et al.'s (2001) obviation principle is often criticized on the basis that sentences such as (7b) and (8b) are well-formed despite containing an overt embedded subject pronoun which does not need to obviate from its closest noun phrase (see also Grano 2015; Huang 2024).

The data discussed by Hu et al. (2001) have sparked ongoing discussion about the possibility of overt subjects in the controlled complement. According to Grano (2015: 147), (9a) and (9b) exemplify copy control, where the overt controllee subjects are optional resumptive pronouns whose optionality is conditioned by the presence vs. absence of an intervening adverbial. Grano (2015: 147) explains that “this kind of conditioning may be consistent with the finding that the acceptability of resumptive pronouns in English *wh*-structures improves with increased syntactic distance between the site of resumption and the landing site (Alexopoulou & Keller 2007; Heestand et al. 2011).” Grano (2015) appears to suggest that the possibility for Mandarin copy control is linked to processing constraints, although he does not elaborate further.

2.3 Exhaustive vs. partial control

Building on data from Hu et al. (2001), Grano (2015) observes that the form of the overt controllee in copy control varies in accordance with the exhaustive control (EC) vs. partial control (PC) properties of the matrix verb. As an illustration, *shefa* ‘try’ is an EC verb, which requires strict identity between the controller and controllee, whereas *dasuan* ‘intend’ is a PC verb, which allows the controller to be a subset of the entities denoted by the controllee. In (10a) and (10b), only the PC verb *dasuan* ‘intend’ but not the EC verb *shefa* ‘try’ allows the collective word *yiqi* ‘together’ in the subordinate clause, signifying partial control. Grano (2015: 148) observes that in (11a) and (11b), the PC verbs *dasuan* ‘intend’ and *qing* ‘invite’ allow overt complement subjects that are partially controlled: Grano (2015) calls this “partial copy control.”

- (10) a. xiaoming_i shefa $\emptyset_{i/\#i+}$ (#yiqi) chi fan
 Xiaoming try $\emptyset$ together eat meal
 ‘Xiaoming tries to have a meal (#together).’ (*shefa* ‘try’ = EC)
- b. xiaoming_i dasuan $\emptyset_{i+}$ yiqi chi fan
 Xiaoming intend $\emptyset$ together eat meal
 ‘Xiaoming intends to have a meal together.’ (*dasuan* ‘intend’ = PC)
- (11) a. wo_i dasuan [tian hei yihou] women_{i+} yiqi qu
 I intend sky dark after 1PL together go
 ‘I intend to go together after it gets dark.’ (Hu et al. 2001: 1132)
- b. wo qing zhangsan_i [jintian xiawu liu dian yihou] ta_i he ta_i taitai yiqi
 I invite Zhangsan today afternoon six o'clock after 3SG and his wife together
 lai chi fan
 come eat meal
 ‘I invited Zhangsan and his wife to come to dinner after six this afternoon.’ (Hu et al. 2001: 1132)

⁷According to Hu et al. (2001: 1134), even when intervening adverbials do not contain an overt NP, “they contain an implicit NP argument since an adverbial phrase can be analyzed into either a subject-oriented or an object-oriented predicate that contains an implicit argument, and this argument is the closest NP that makes the matrix subject in question less close and less prominent and hence can be the antecedent of the overt pronoun.”

In related work, Li (2024) observes that, when presented with examples containing overt controllees, her consultants tend to prefer null controllees; however, Li (2024) claims that copy control is still considered possible by her consultants under PC verbs but not EC verbs. What makes Li's (2024) investigation stand out from previous studies is her careful differentiation between PC and EC verbs in their copy-control licensing capabilities. Notably, different from Hu et al. (2001) and Grano (2015), Li (2024) claims that no intervening adverbial is required for copy control under PC verbs, and that examples like (12) are acceptable. This reflects a need to test for the necessity of the intervening adverbial.

- (12) lisi dasuan ta bu yao qu shang daxue
 Lisi intend 3SG not will go attend college
 'Lisi intends not to attend college.' (Li 2024: 340, her judgment)

Li's (2024) discussion centres on four verbs, namely *shefa* 'try', *dasuan* 'intend', *quan* 'persuade', and *rang* 'let', each of which demonstrates a different control property. Her argument that there is a contrast between EC and PC verbs in copy control relies on her comparison between *shefa* 'try' (EC) and *dasuan* 'intend' (PC). Ideally, more EC and PC verbs are required to substantiate the empirical generalizations. In Section 3, we present an experimental paradigm to systematically test for the necessity of the intervening adverbial as well as PC vs. EC contrasts.

2.4 Form of the overt controllee

Zhang (2016) identifies three types of overt controllees: complex complemented pronouns, which comprise a pronoun, numeral, classifier, and head noun; bound lexical pronouns; and the reflexive pronoun *ziji* 'self'. Examples (1), (9b), and (11a) involve lexical pronouns as controllees which are (partially) bound by their controllers. In (9a), *wo yi-ge ren* 'I one-CL person' is a complex complemented pronoun. As Zhang (2016) observes, complex complemented pronouns and reflexive *ziji* can serve as either adverbials or arguments, unlike (bound) lexical pronouns, which do not have an adverbial use. To differentiate between the two functions, she proposes using the focus markers *lian...dou* as a diagnostic to rule out the adverbial use. If a complex complemented pronoun or reflexive in a controlled complement can occur with these focus markers, it is classified as an argument and, therefore, an overt controllee; if not, it is an adverbial.

- (13) a. [baba he mama]_i dasuan wanshang lian [tamen liang ge ren/ ziji]_i dou qu duchang
 dad and mom plan evening even 3PL two CL person/ self also go casino
 'Dad and mom made the plan that even they two go to a casino this evening.' (Zhang 2016: 287, her judgment)
- b. *lili_i lian [ta yi ge ren/ ziji]_i dou qu-le duchang
 Lili even 3SG one CL person/ self also go-PRF casino
 'Even Lili went to a casino by herself.' (Zhang 2016: 281, her judgment)

Zhang (2016) judges example (13a) as grammatical; thus, according to her focus-marker diagnostic, the complex complemented pronoun and reflexive *ziji* in the controlled complement of *dasuan* 'plan' should be classified as overt controllees. In contrast, the complex complemented pronoun and reflexive in example (13b) cannot appear with focus markers, classifying them as adverbial. However, the grammaticality of (13a) is disputed. Yan (2022: 39) reports that the sentence is judged ungrammatical by some native speakers. This suggests that the controllee status of complex complemented pronouns and the reflexive *ziji* is unclear, due to differing native-speaker judgments involving the focus-marker diagnostic. Therefore, to identify instances of copy control, we maintain that the crucial evidence comes

from examples with bound lexical pronouns as the overt controllee (or as part of the overt controllee such as (11b), which involves partial copy control).

2.5 Gradient acceptability and acceptability-judgment experiments

Ussery et al. (2016), Paul (2018), Huang (2022), He (2020), and Huang (2024) have expressed concerns about the grammaticality of Mandarin examples claimed to involve copy control. Huang (2024) is the first study to employ acceptability-judgment experiments to evaluate Mandarin copy control. Overall, he concludes that an overt controllee reduces the acceptability of control constructions, and he therefore concludes that Mandarin shows a classical finiteness distinction, where control verbs select for non-finite complements that do not license overt subjects.

One issue for Huang (2024) is that his experimental stimuli heavily rely on complex complemented pronouns. As discussed above, whether a complex complemented pronoun can truly serve as the overt controllee in a controlled complement is controversial. In fact, based on a series of diagnostics, Li (2024: 344–347) classifies complex complemented pronouns occurring under EC verbs, such as those in (14c) and (14d)—which form a significant portion of Huang’s (2024) experimental stimuli—as adverbials, rather than overt subject controllees as claimed by Huang (2024).

- (14) a. “Null subject, no adverbial” condition (Huang 2024: 540)
zhe-ge qiezei shefa taochu juliu-suo
this-CL burglar try escape detention-centre
‘This burglar tried to escape from the detention center.’
b. “Null subject, adverbial” condition (Huang 2024: 540)
zhe-ge qiezei shefa [wuyue shifen] taochu juliu-suo
this-CL burglar try midnight time escape detention-centre
‘This burglar tried to escape from the detention center at midnight.’
c. “Overt subject, no adverbial” condition (Huang 2024: 540)
zhe-ge qiezei shefa ta yi-ge ren taochu juliu-suo
this-CL burglar try 3SG one-CL person escape detention-centre
‘This burglar tried to escape from the detention center by himself.’
d. “Overt subject, adverbial” condition (Huang 2024: 540)
zhe-ge qiezei shefa [wuyue shifen] ta yi-ge ren taochu juliu-suo
this-CL burglar try [midnight time] 3SG one-CL person escape detention-centre
‘This burglar tried to escape from the detention center by himself at midnight.’

Huang’s experimental analysis also raises the issue of accounting for different degrees of acceptability. (14a)–(14d) exemplify the conditions in one of Huang’s (2024) experiments, where (14a) and (14b) are the baseline conditions without copy control. These were, as expected, rated with high scores. According to Huang (2024: 541), if the “overt subject, adverbial” sentences are grammatical, we should “expect these sentences to have a mean acceptability comparable to the two null subject conditions, given their similarities in structure and meaning.” While we sympathize with this criterion, we have reservations about whether copy control sentences should be expected to show the same level of acceptability as the corresponding examples without an overt subject. In fact, it has been noted in cross-linguistic investigation that copy control can sometimes give rise to a sense of redundancy. In the context of Telugu copy control, mentioned in Section 1, Haddad (2009: 84) observed that some copy-control sentences “were judged by my Telugu consultants as redundant, but not unacceptable. According to them, pronouncing only one of the subjects is sufficient. In other words, Forward or Backward Control can do the job.” From a functionalist perspective, it is not surprising that speakers may prefer to use fewer linguistic elements to convey the same meaning (see, e.g., Givón 1984; Haiman 1983). It is known that participants’ acceptability judgements are influenced by multiple factors, with grammaticality being

only one of them (see, e.g., Goodall 2021), and we conjecture that this sense of redundancy may reduce participants’ ratings in acceptability experiments in comparison to non-copy-control sentences.

Interestingly, Haddad’s (2009: 84) research on Telugu found that the “redundancy is eliminated...if the sentence is made longer” by intervening adverbial(s) between the controller and overt controllee, as shown in (15), with *pooyina-waaram* ‘last week’ as an intervening adverbial. This observation is strikingly similar to the claims by Hu et al. (2001) and Grano (2015) about the significance of intervening adverbials in Mandarin copy control. We will return to the issue of intervening adverbials in Mandarin copy control in Section 5.

(15) Telugu (Haddad 2009: 84)

[(Kumaar-ki_i) pooyina-waaram jwaram wacc-i] (atanu_i/ Kumaar_i) muuDu roojulu
Kumar-DAT last-week fever.NOM come-PART.CNP (he/ Kumar.NOM) three days
 mandulu waaDaa-Du
 medicines used-3.M.S
 ‘Kumar had a fever last week, and he/Kumar was on medication for three days.’

Returning to Huang’s (2024) conditions, a comparison between (14c) and (14d) would shed light on the licensing conditions of copy control and the significance of the intervening adverbial, as proposed by Hu et al. (2001) and Grano (2015). If there is any statistically significant difference between the judgments of (14c) and (14d), its theoretical implications will need to be articulated. Instead of directly comparing (14c) and (14d), Huang (2024) uses a “differences in differences” (DD) score, which calculates whether the acceptability difference between (14c) and (14d) is greater than that between (14a) and (14b). Although his results do show a statistically significant difference in the DD score (Huang 2024: 544), no theoretical explanation is offered regarding what causes the difference. The lack of explanation might well be, as we conjecture, due to the fact that the copy-control conditions overall receive lower scores than the non-copy-control ones (Huang 2024: 543)—an observation which aligns with Haddad’s (2009) fieldwork experience of Telugu. Overall, a plausible conclusion, following Grano (2015), is to acknowledge the need to include copy control and its licensing conditions in Mandarin grammar.

3 Empirical patterns of Mandarin copy control

In this section, we will address the following two research questions:

1. What are the empirical patterns of copy control in Mandarin Chinese?
2. How does copy control interact with other control properties in Mandarin (including exhaustive control, partial control, raising, and transitive subject control)?

To address question (1), we will examine empirical patterns in copy control via acceptability experiments. Regarding question (2), we include raising verbs in our investigation because LFG assimilates raising verbs into its control theory: raising verbs are the canonical exemplars of what LFG calls *functional control*, while control verbs can instantiate either *anaphoric control* or *functional control*, as will be discussed in Section 4.1.3. The distinction between anaphoric and functional control will be important for our analysis.

We differentiate between control verbs and raising verbs based on theta-role tests featured in the general syntactic literature (e.g., Adger 2003; Radford 2016). Verbs such as *sihu* ‘seem’ and *haoxiang* ‘appear’ are raising verbs subcategorizing for an athematic subject. Following Lam (2025: 225), we identify raising verbs via the semantic restriction test in (16), contrasting the raising verbs *sihu* ‘seem’ and *haoxiang* ‘appear’ with the control verbs *xiangbanfa* ‘try’ and *dasuan* ‘intend’:

- (16) a. *shiqing* *sihu*/*haoxiang* *fasheng-le*
 event seem/appear happen-PRF
 ‘The event seems/appears to have happened.’
 b. *#shiqing* *xiangbanfa*/*dasuan* *fasheng-le*
 event try/intend happen-PRF
 #‘The event tries/intends to have happened.’

Xiangbanfa ‘try’ and *dasuan* ‘intend’ impose semantic restrictions on their subjects, which must be sentient participants capable of trying or intending. As *shiqing* ‘event’ does not fulfil these requirements, the examples in (16b) are semantically anomalous. In contrast, *sihu* ‘seem’ and *haoxiang* ‘appear’ do not impose such semantic restrictions on their subjects; instead, they assign theta roles to the embedded proposition, with *shiqing* ‘event’ being part of the embedded proposition. For further tests distinguishing control and raising verbs in Mandarin, including tests involving weather verbs, see Lam (2023: 33-39).

We will also investigate how copy control interacts with a less-studied control phenomenon pertinent to verbs that subcategorize for an object but enforce subject control, similar to the English verb *promise*. Based on introspective judgment, Lam (2023) suggests that some of these transitive subject-control verbs, such as *daibiao* ‘represent/act on behalf’ and *xue* ‘learn from’, cannot license copy control. These verbs appear to be quite Mandarin-specific with no direct English equivalents. Evidence for their status as control verbs includes the obligatoriness of the embedded clause, which indicates that the embedded clause is one of the subcategorized arguments, as well as the exhaustive reading of the controllee, suggesting EC properties:

- (17) a. *yuehan_i* *daibiao* [*quan ban tongxue*]_j *($\emptyset_i/\#i+/*j$) (*#yiqi*) *huijian zongtong*
 John represent whole class classmate $\emptyset$ together meet President
 Lit: ‘John represented the whole class of classmates to meet the President (#together).’
 ‘John met the President on behalf of the class.’
 b. *yuehan_i* *xue* *luxi_j* *($\emptyset_i/\#i+/*j$) (*#yiqi*) *changge*
 John learn.from Lucy $\emptyset$ together sing
 Lit: ‘John learned from Lucy to sing (#together).’
 ‘John learned to sing from Lucy.’

We will compare these verbs with transitive subject-control verbs exhibiting PC properties, such as *daying* ‘promise’ and *konghe* ‘threaten’, to investigate whether there is any correlation between these two classes of transitive subject-control verbs and the possibility of copy control.

- (18) *yuehan_i* *dayin/konghe* *bite_j* *($\emptyset_i/i+/*j$) (*yiqi*) *huijian xiaozhang*
 John promise/threaten Peter $\emptyset$ together meet principal
 ‘John promised/threatened Peter to meet the principal (together).’

Before proceeding to our experimental study, we provide a list of the types of verbs that will be relevant in our discussion:

- (19) Exhaustive control (EC)
 The controller and controllee of an EC verb must be identical in reference.
 Example verbs: *shefa*/*xiangbanfa*/*changshi* ‘try’, *deyi* ‘manage’

- (20) Partial control (PC)
The controller of a PC verb may refer to a subset of the entities denoted by the controllee.
Example verbs: *dasuan* ‘intend’, *zhunbei* ‘prepare’
- (21) Raising
A raising verb subcategorizes for an athematic grammatical relation which is also the subject of a complement clause.
Example verbs: *sihu* ‘seem’, *haoxiang* ‘appear’
- (22) Transitive subject-control verbs
A transitive subject-control verb subcategorizes for an object, but the controller is the subject rather than the object.
Example verbs: *daibiao* ‘represent’, *konghe* ‘threaten’

In the following sections, we present the design and results of two acceptability experiments, and summarize the empirical generalizations regarding copy control which emerge from these experiments.

3.1 Acceptability-judgment experiments: Experimental design

We conducted two acceptability-judgment experiments. In light of the discussion in Section 2.4, we used only bound lexical pronouns as overt controllees. Both experiments involved rating sentences on a 7-point Likert scale. The Supplementary Materials document contains further details.

3.1.1 Experiment 1: Intervening adverbial and overt controllee in copy control and raising

Experiment 1 investigates whether and how the possibility for an overt controllee is affected by control properties (EC, PC, and raising) and the presence (or absence) of an intervening adverbial (ADV). The experiment followed a 4x2 factorial design, as shown in Table 1, generating eight conditions, illustrated by the examples in (23). “PC, equal coreference” refers to the situation where a partial control verb is used but the overt controllee is a singular pronoun bound by a singular matrix subject; “PC, partial coreference” refers to the situation where a partial control verb is used and the overt controllee is a plural pronoun partially bound by a singular matrix subject. Since there has always been consensus that control and raising sentences without an overt controllee are grammatically well-formed, and the experimental results of Huang (2024) have again proved this point, in order to simplify the experiment and reduce the number of testing conditions, Experiment 1 focused on copy control and raising with overt pronominal subordinate subjects, and we were interested in potential variation in judgements of copy control and raising based on control properties and an intervening adverbial.

	EC	PC, equal coref.	PC, partial coref.	Raising
Intervening ADV	Condition A	Condition C	Condition E	Condition G
No intervening ADV	Condition B	Condition D	Condition F	Condition H

Table 1: Conditions of Experiment 1.

- (23) a. *Condition A: EC, intervening ADV*
 zhe-ming waijian.guan xiangbanfu [zai bi-men huiyi jinxing qijiang] ta
 this-CL diplomat try at closed-door meeting on.going during 3SG
 huijian daibiao.tuan tuan.zhang
 meet delegation leader
 ‘This diplomat tried to meet the leader of the delegation during the closed-door meeting.’
- b. *Condition B: EC, no intervening ADV*
 zhe-ming waijian.guan xiangbanfu ta huijian daibiao.tuan tuan.zhang
 this-CL diplomat try 3SG meet delegation leader
 ‘This diplomat tried to meet the leader of the delegation.’
- c. *Condition C: PC (equal coreference), intervening ADV*
 zhe-ming waijian.guan dasuan [zai bi-men huiyi jinxing qijiang] ta huijian
 this-CL diplomat intend at closed-door meeting on.going during 3SG meet
 daibiao.tuan tuan.zhang
 delegation leader
 ‘This diplomat intended to meet the leader of the delegation during the closed-door meeting.’
- d. *Condition D: PC (equal coreference), no intervening ADV*
 zhe-ming waijian.guan dasuan ta huijian daibiao.tuan tuan.zhang
 this-CL diplomat intend 3SG meet delegation leader
 ‘This diplomat intended to meet the leader of the delegation.’
- e. *Condition E: PC (partial coreference), intervening ADV*
 zhe-ming waijian.guan dasuan [zai bi-men huiyi jinxing qijiang] tamen
 this-CL diplomat intend at closed-door meeting on.going during 3PL
 yiqi huijian daibiao.tuan tuan.zhang
 together meet delegation leader
 ‘This diplomat intended to meet the leader of the delegation together (with someone) during the closed-door meeting.’
- f. *Condition F: PC (partial coreference), no intervening ADV*
 zhe-ming waijian.guan dasuan tamen yiqi huijian daibiao.tuan tuan.zhang
 this-CL diplomat intend 3PL together meet delegation leader
 ‘This diplomat intended to meet the leader of the delegation together (with someone).’
- g. *Condition G: Raising, intervening ADV*
 zhe-ming waijian.guan sihu [zai bi-men huiyi jinxing qijiang] ta huijian
 this-CL diplomat seem at closed-door meeting on.going during 3SG meet
 daibiao.tuan tuan.zhang
 delegation leader
 ‘This diplomat seemed to meet the leader of the delegation during the closed-door meeting.’
- h. *Condition H: Raising, no intervening ADV*
 zhe-ming waijian.guan sihu ta huijian daibiao.tuan tuan.zhang
 this-CL diplomat seem 3SG meet delegation leader
 ‘This diplomat seemed to meet with the leader of the delegation.’

Eight sentence frames with distinct lexicalizations were paired with three groups of verbs according to the eight experimental conditions: (i) EC verbs: *xiangbanfa* ‘try’ and *deyi* ‘manage’; (ii) PC verbs: *dasuan* ‘intend’, *zhunbei* ‘prepare’, *yuanyi* ‘be willing’, and *zhiyi* ‘insist’; (iii) raising verbs: *sihu* ‘seem’

and *haoxiang* ‘appear’. The categorization of these verbs as EC control, PC control, and raising has been established in the existing literature: see Grano (2015) and Lam (2023). In total, there were 64 testing sentences distributed across eight lists in a Latin Square design. In every list, each of the eight sentence frames appeared once, and each verb appeared once in its allocated sentence frame.

3.1.2 Experiment 2: Transitive subject-control verbs and copy control

Experiment 2 compared two types of transitive subject-control verbs. As noted in Section 2, based on introspection, Lam (2023) posits that transitive subject-control verbs enforcing exhaustive control, such as *daibiao* ‘represent/act on behalf’ and *xue* ‘learn from’, cannot license copy control. We compare them with another group of transitive subject-control verbs allowing partial control, including *daying* ‘promise’ and *konghe* ‘threaten’. By comparing these EC vs. PC transitive subject-control verbs in copy-control environments, our experiment tested whether some or all transitive subject-control verbs allow copy control. Table 2 lists the experimental conditions, and (24) exemplifies these conditions. All these conditions involved copy control, again given the well-established consensus that control sentences without an overt controllee are grammatically well-formed. We created four sentence frames of distinct lexicalization for Experiment 2 paired with two groups of transitive subject-control verbs according to the four experimental conditions: (i) transitive subject-control verbs allowing PC: *daying* ‘promise’ and *konghe* ‘threaten’; (ii) transitive subject-control verbs requiring EC: *daibiao* ‘represent/act on behalf’ and *xue* ‘learn from’. There were 16 testing sentences distributed across four lists in a Latin Square design. In every list, each of the four sentence frames and each verb appeared once.

	Transitive subject control allowing PC (e.g., <i>daying</i> ‘promise’)	Transitive subject control enforcing EC (e.g., <i>daibiao</i> ‘represent’)
Intervening ADV	Condition A	Condition C
No intervening ADV	Condition B	Condition D

Table 2: Conditions of Experiment 2.

- (24) a. *Condition A: Transitive subject control allowing PC, intervening ADV*
zhe-ming tanpan.guan daying difang [zai zuihou qixian daolai zhiqian] ta
this-CL negotiator promise enemy.PL at final deadline arrive before 3SG
qianshu tinghuo xieyi
sign ceasefire agreement
‘The negotiator promised the enemy to sign the ceasefire agreement before the final deadline.’
- b. *Condition B: Transitive subject control allowing PC, no intervening ADV*
zhe-ming tanpan.guan daying difang ta qianshu tinghuo xieyi
this-CL negotiator promise enemy.PL 3SG sign ceasefire agreement
‘The negotiator promised the enemy to sign a ceasefire agreement.’
- c. *Condition C: Transitive subject control enforcing EC, intervening ADV*
zhe-ming tanpan.guan daibiao difang [zai zuihou qixian daolai zhiqian] ta
this-CL negotiator represent enemy.PL at final deadline arrive before 3SG
qianshu tinghuo xieyi
sign ceasefire agreement
Lit: ‘The negotiator represented the enemy to sign the ceasefire agreement before the final deadline.’
‘The negotiator, on behalf of the enemy, signed a ceasefire agreement before the deadline.’

- d. *Condition D: Transitive subject control enforcing EC, no intervening ADV*
 zhe-ming tanpan.guan daibiao difang ta qianshu tinghuo xieyi
 this-CL negotiator represent enemy.PL 3SG sign ceasefire agreement
 Lit: ‘The negotiator represented the enemy to sign a ceasefire agreement.’
 ‘The negotiator, on behalf of the enemy, signed a ceasefire agreement.’

Note that *daibiao* ‘promise’, like its English equivalent, is capable of control shift, although the subject-control reading is preferred (Landau 2000). To ensure that we tested the subject-control property of this verb, our experimental stimuli used a singular matrix subject and a singular lexical pronoun as the controllee together with a plurality-denoting matrix object, so that the matrix subject is the only suitable antecedent for the controllee pronoun.⁸

3.2 Acceptability-judgment experiments: Participants and apparatus

The acceptability-judgment experiments were distributed as questionnaires on Qualtrics. Each questionnaire contained 12 experimental stimuli from the two experiments (eight from Experiment 1; four from Experiment 2); they were interspersed with 20 fillers of comparable syntactic complexity, displaying different degrees of acceptability.⁹ The sentences were presented in a randomized fashion. Among the set of fillers were five sentences of high acceptability and five which were highly unacceptable, forming “gold-standard” fillers that helped spot invalid responses. We established these “gold-standard” fillers based on a pilot run with native-speaker consultants.

In total, 83 participants submitted their responses. Clear instructions and examples were provided before the rating process began. Participants were asked about their language background, including when they started acquiring Mandarin, a self-report of their language competence, etc. We only included responses from participants who started acquiring Mandarin before age 6. Participants were instructed to rate sentences based on a 7-point Likert scale, with a score of 1 indicating a completely unacceptable sentence, and 7 indicating a perfectly natural and acceptable sentence. Participants were further screened based on their ratings on the “gold-standard” fillers. If a participant rated two or more of the highly acceptable fillers with scores below 4, their response was discarded; likewise, if they rated two or more of the highly unacceptable fillers with scores above 4, their response was discarded. After the screening procedures, responses from 59 participants were considered valid and used in the data analysis.¹⁰

⁸There are further complications concerning *daibiao* ‘promise’, particularly its ability to function as a non-control verb, where the embedded subject can freely refer to any discourse-salient entity (see Section 2). Nevertheless, for this non-control structure to sound natural, native speakers typically require the presence of a modal verb such as *hui* ‘will’ or *keyi* ‘can’ in the complement clause, as illustrated in (25).

(25) yuehan_i dayin bite_j tai_{i/j} #(hui/keyi) huijian xiaozhang
 John promise Peter 3SG will/can meet principal

‘John promised Peter that he will/can meet the principal.’

In some analyses, *hui* ‘will’ is treated as a finiteness marker (e.g., Huang 1989); thus, its presence signals that the complement clause is finite and non-control (cf. Hu et al. 2001; Lam 2025 for alternative accounts). Because an additional modal is generally needed for *daibiao* ‘promise’ to receive a non-control interpretation, we assume that *daibiao* ‘promise’ is a subject-control verb when the complement does not contain a modal.

⁹We follow past studies of experimental syntax, e.g., Huang (2021), where combining experimental stimuli from different experiments is not uncommon, partly driven by operational needs.

¹⁰Another way of screening participants that has been used in other studies (e.g., Huang 2024) is to calculate the group means of the fillers and discard participants whose responses have a certain number of extremes which are two standard deviations above or below the group mean. We tried this method and noticed that fewer participants with unreliable responses would have been discarded. The criteria we set for our study were thus stricter. We preferred to stay with the stricter criteria to ensure that the responses are trustworthy.

3.3 Experimental results, statistical analysis, and empirical generalizations

The results of the two experiments are presented in boxplots created by the tool BoxPlotR (Spitzer et al. 2014). The notches represent the 95% confidence intervals of the medians. The black crosses mark mean ratings. The gray areas around the crosses indicate the 95% confidence intervals of the means. Raw acceptability scores (1–7) are shown in the boxplots. The scores provide approximate ideas about how good the sentences were to the participants. On a Likert scale of 1 to 7, we expect that 6–7 points indicates high acceptability; 5 is still a positive response; 4 suggests neutrality; 3 indicates a negative response; 1–2 suggests (very) low acceptance. The acceptability scores are analyzed for statistical significance. Z-score transformation is applied to each participant to remove any bias in the use of the Likert scale. A linear mixed-effects model is constructed for each experiment using the R packages `lme4` and `lmerTest` (Bates et al. 2015; Kuznetsova et al. 2013). The z-scored ratings are the dependent variable. This design for statistical analysis follows Huang (2024) and many others in experimental syntax (e.g., Bross 2019; Huang 2021; Huang & Phillips 2021; Juzek 2015; Lam 2024; Sprouse et al. 2016).

3.3.1 Results of Experiment 1

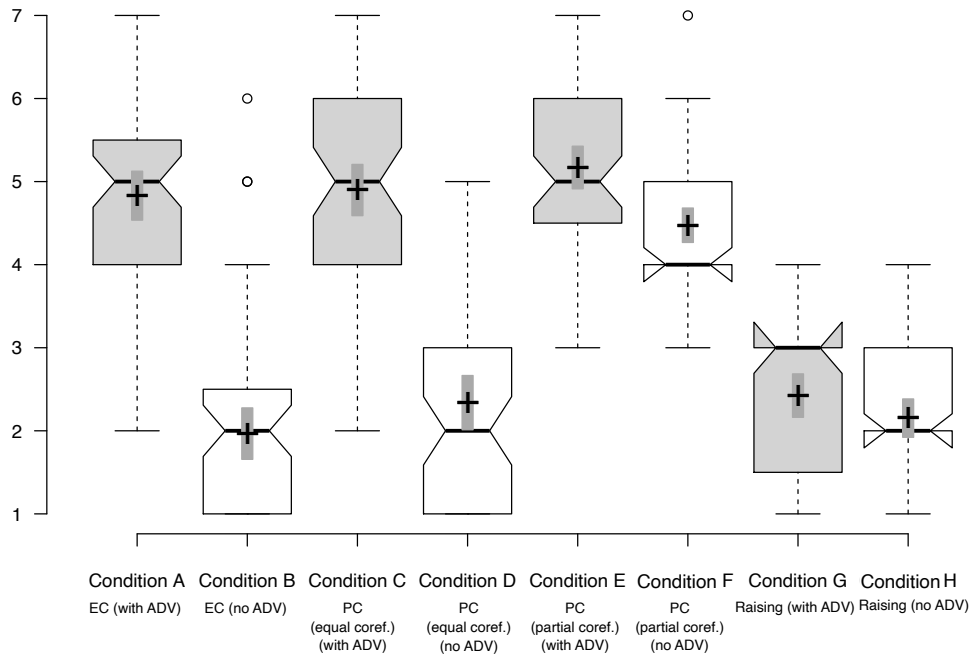


Figure 1: Acceptability scores of Experiment 1

Figure 1 presents the results of Experiment 1, which investigated how copy control is affected by control properties (EC, PC, and raising) and the presence of an intervening adverbial (ADV). All the conditions involved the presence of an overt pronoun as the subordinate-clause subject. Based on whether the notches overlap, we can estimate whether there are likely to be significant differences between the conditions. Comparing Conditions A and B as well as Conditions C and D, the presence of an intervening adverbial contributed to a substantial improvement in acceptability for EC sentences and PC sentences with equal coreference. Although the lower and upper whiskers (Tukey-style) of Conditions A and C are at point 2 and 7, suggesting interspeaker variation, their interquartile ranges are both above point 4 and their medians are at point 5, suggesting overall positive responses. This appears to echo previous

accounts of copy-control structures (e.g., Haddad 2009; Li 2024): although speakers may have a feeling of redundancy about the overtly expressed controllee, they still consider the structures to be acceptable. In contrast, Conditions B and D show poor acceptability with a median at point 2. Regarding Conditions E and F, which involve PC verbs with partial coreference, because their notches do not overlap, we expect a significant difference between them; nevertheless, the extent of improvement caused by the presence of an intervening adverbial appears to be less than the previous conditions. The median of Condition E is 5 and that of Condition F is 4. Comparing Conditions D and F, although they both involve PC verbs with no intervening adverbial, a marked improvement is noted when there is partial coreference between the controller and controllee. Most participants rated Condition F with a score between 4 and 6. As for Conditions G and H, the fact that all of the ratings are below 4 (with medians of 3 and 2 respectively) indicates low levels of acceptance for raising verbs with an overt controllee, regardless of the intervening adverbial.¹¹

The linear mixed-effects model for Experiment 1 features control properties and presence (or absence) of the intervening adverbial as the fixed effects; the participants, sentence frames, and verb choices are the random effects to control for idiosyncratic effects they might exert on acceptability ratings, similar to Huang’s (2024) design.¹²

Levels	Estimate	Std. error	t-value	p
<i>(EC, no intervening ADV as reference level)</i>				
PC (equal coreference)	0.17	0.12	1.50	0.16
PC (partial coreference)	1.18	0.12	10.29	<0.001
Raising	0.08	0.12	0.61	0.56
Intervening ADV	1.37	0.08	16.67	<0.001

Table 3: Linear mixed-effects modeling analysis for Experiment 1.

Contrasts	Estimate	SE	t-ratio	p
<i>(No-intervening-ADV conditions)</i>				
EC – PC (partial coreference)	-1.18	0.12	-10.28	<0.001
PC (equal coreference) – PC (partial coreference)	-1.01	0.08	-12.30	<0.001
PC (partial coreference) – Raising	1.11	0.12	9.62	<0.001
<i>(Intervening ADV conditions)</i>				
EC – Raising	1.17	0.12	9.45	<0.001
PC (equal coreference) – Raising	1.21	0.12	10.49	<0.001
PC (partial coreference) – Raising	1.33	0.12	11.57	<0.001
<i>(EC conditions)</i>				
ADV – No ADV	1.37	0.08	16.65	<0.001
<i>(PC with equal coreference conditions)</i>				
ADV – No ADV	1.23	0.08	14.98	<0.001
<i>(PC with partial coreference conditions)</i>				
ADV – No ADV	0.35	0.08	4.21	<0.001

Table 4: Contrasts for Experiment 1.

Table 3 shows the results of the fixed effects when EC (no intervening ADV) is set as the reference level. An ANOVA analysis of the linear mixed-effects model for Experiment 1 further reveals interaction between control properties and adverbial ($F(3, 460.25) = 57.91, p < 0.001$). To show the contrasts

¹¹Although the notches do not overlap, after z-transformation, our linear mixed-effects modeling analysis suggests that there is no statistically significant difference between Conditions G and H ($p > 0.05$).

¹²Our formula is $z\text{-score} \sim \text{control} * \text{adverbial} + (1 | \text{participant}) + (1 | \text{frame}) + (1 | \text{verb})$.

between the levels more clearly, we use the `emmeans` package (Lenth 2017) to derive pairwise comparisons. Table 4 presents those contrasts which are statistically significant and relevant to our discussion. The results corroborate our previous discussion based on the descriptive statistics revealed by the boxplot. As shown in Table 4, when there is no intervening adverbial, PC (partial coreference) conditions score much higher than all other control and raising conditions (absolute differences in estimates > 1 , $p < 0.001$). Regarding the conditions with an intervening adverbial, all other control conditions score much higher than the raising condition (differences in estimates > 1 , $p < 0.001$). For the EC and PC (equal coreference) conditions, the presence of an intervening adverbial substantially improves the acceptability (differences in estimates > 1 , $p < 0.001$). Although similar improvement is witnessed for PC (partial coreference) conditions, the extent of improvement is much smaller (difference in estimates = 0.35, $p < 0.001$).

Based on Experiment 1, we propose the following empirical generalizations:

(26) Empirical Generalization One

For EC- and PC-verb constructions where the controller and controllee share identical reference, copy control must be licensed by the presence of an intervening adverbial.

(27) Empirical Generalization Two

For PC-verb constructions where the controller is a subset of the entities denoted by the controllee, copy control is possible, regardless of the presence (or absence) of an intervening adverbial.

(28) Empirical Generalization Three

Raising constructions do not allow an overt embedded subject, regardless of the presence (or absence) of an intervening adverbial.

3.3.2 Results of Experiment 2

Experiment 2 compared two types of transitive subject-control verbs, one allowing partial control and the other enforcing exhaustive control, with respect to their copy-control behaviour. Figure 2 presents the results of Experiment 2 using a boxplot. The lower and upper whiskers of Condition A (verbs allowing PC, with intervening adverbial) demonstrate a wide range of interspeaker variation, with ratings varying from 1 to 7. That being said, the interquartile range is confined to the values between 2 and 4, and the median is 3, which indicates negative impression. For Conditions B, C, and D, their medians are at point 2 and their upper whisker is at point 3, both of which suggest poor acceptability.

The linear mixed-effects model for Experiment 2 features EC vs. PC transitive subject control and presence (or absence) of the intervening adverbial as the fixed effects; the participants, sentence frames, and verb choices are the random effects to control for idiosyncratic effects, similar to Huang’s (2024) design.¹³ Table 5 shows the results of the model when verbs enforcing EC (no intervening ADV) is set as the reference level. A post hoc ANOVA analysis of the linear mixed-effects model reveals that there is no significant interaction between transitive subject control types and the intervening adverbial ($F(1, 228.37) = 2.42$, $p = 0.12$). To display the contrasts between the different levels, the `emmeans` package (Lenth 2017) is used to produce pairwise comparison shown in Table 6. The only statistically significant contrast is between the presence and absence of the intervening adverbial for transitive subject-control verbs allowing PC, but the effect size is small (difference in estimates = -0.24, $p < 0.01$).

¹³Our formula is $z\text{-score} \sim \text{control} * \text{adverbial} + (1 | \text{participant}) + (1 | \text{frame}) + (1 | \text{verb})$.

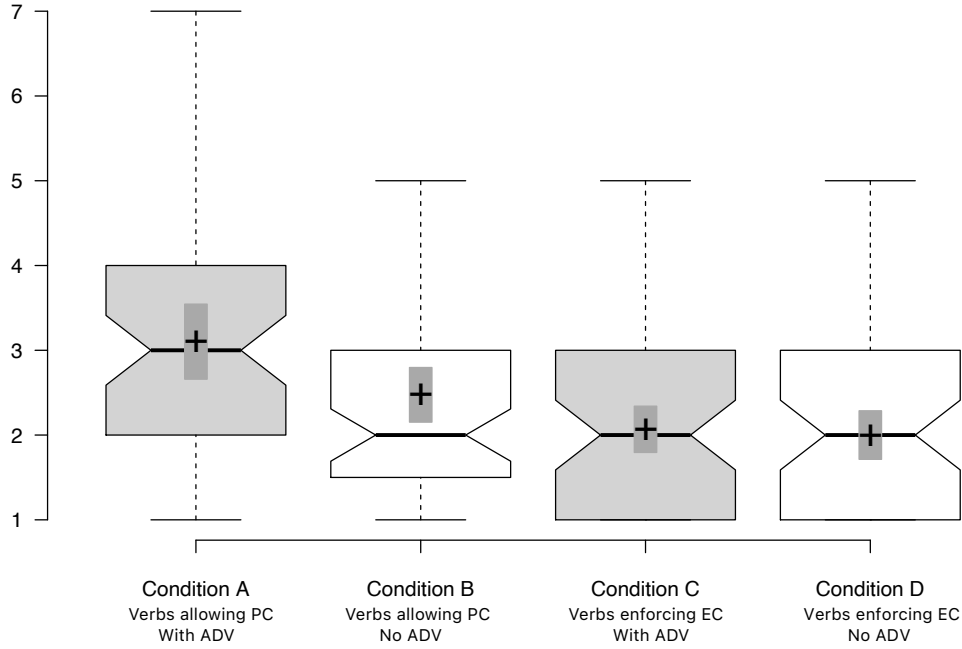


Figure 2: Acceptability scores of Experiment 2.

Levels	Estimate	Std. error	t-value	p
<i>(Verbs enforcing EC, no intervening ADV as reference)</i>				
Verbs allowing PC	0.26	0.24	1.08	0.42
Intervening ADV	0.038	0.09	0.42	0.67

Table 5: Linear mixed-effects modeling analysis for Experiment 2.

After reviewing the boxplot statistics and the results of the mixed-effects model, we are not convinced that transitive subject-control verbs are capable of licensing copy control, regardless of the presence of the intervening adverbial. Our conclusion thus supports the introspective judgment of Lam (2023). Admittedly, there is interspeaker variation regarding the acceptability of constructions with transitive subject-control verbs allowing PC and an intervening adverbial, as noted in the boxplot statistics; nevertheless, the median is only 3, and the mixed-effects model shows only minimal effect regarding the adverbial (difference in estimates = -0.24). In fact, based on Experiment 1, we would expect verbs allowing PC to be more amenable to copy control; the results of Experiment 2 thus also indicate that transitive subject-control verbs resist copy control regardless of their EC/PC properties. We posit the following generalization:

(29) Empirical Generalization Four

Transitive subject-control verbs cannot license copy control.

3.4 Interim summary

Based on our experimental results, we have derived four empirical generalizations of how copy control interacts with other control properties in Mandarin Chinese, repeated as follows:

Contrasts	Estimate	SE	t-ratio	p
<i>(No-intervening-ADV conditions)</i>				
Verbs allowing PC – Verbs enforcing EC	0.26	0.24	1.079	0.41
<i>(Intervening ADV conditions)</i>				
Verbs allowing PC – Verbs enforcing EC	0.46	0.24	1.91	0.21
<i>(Verbs-allowing-PC conditions)</i>				
No ADV – ADV	-0.24	0.09	-2.62	< 0.01
<i>(Verbs-enforcing-EC conditions)</i>				
No ADV – ADV	-0.39	0.09	-0.42	0.68

Table 6: Contrasts for Experiment 2.

(30) Empirical Generalization One

For EC- and PC-verb constructions where the controller and controllee share identical reference, copy control must be licensed by the presence of an intervening adverbial.

(31) Empirical Generalization Two

For PC-verb constructions where the controller is a subset of the entities denoted by the controllee, copy control is possible, regardless of the presence (or absence) of an intervening adverbial.

(32) Empirical Generalization Three

Raising constructions do not allow an overt embedded subject, regardless of the presence (or absence) of an intervening adverbial.

(33) Empirical Generalization Four

Transitive subject-control verbs cannot license copy control.

Our results demonstrate the availability of copy control in Mandarin; this finding is consistent with Grano (2015); Hu et al. (2001); Li (2024); Zhang (2016), etc., but stands in contrast to He (2020); Huang (2024); Paul (2018); Ussery et al. (2016), etc., discussed in Section 2. While diverging from Li (2024) but aligning with Hu et al. (2001) and Grano (2015), our findings highlight the essential role of the intervening adverbial in licensing copy control in EC and PC (equal coreference) constructions. Furthermore, we have shed new light on different control types, including the substantial difference between EC (equal coreference) and PC (partial coreference) constructions in their ability to license copy control, the fact that raising constructions do not allow overt controllees, and the fact that transitive subject-control verbs forbid copy control.

4 Formal account of copy control and its interaction with other control properties in Mandarin

In this section, we address the following questions:

1. What are the implications of the empirical patterns? What role does copy control play in the grammar of Mandarin?
2. How can we formally model Mandarin copy control, taking into account its interaction with other control properties?

As discussed in Section 2.1, most previous studies addressing these patterns assume that Mandarin distinguishes finite from non-finite subordinate clauses. Studies that reject copy control, such as Huang (1989) and Huang (2024), posit that Mandarin illustrates a classical finiteness distinction, where control verbs subcategorize for non-finite complements that disallow overt subjects. Zhang (2016) advocates another approach in which Mandarin patterns with languages such as Hungarian, where overt controllees can appear in non-finite environments—contrary to predictions made by the classical finiteness account. While Huang (2024) and Zhang (2016) assume that both EC and PC verbs select for non-finite complements, Li (2024) proposes that EC verbs select for non-finite complements and PC verbs select for finite ones. In contrast, for Hu et al. (2001), all so-called control verbs allow overt subjects under certain licensing conditions, and Mandarin does not make any finiteness distinctions.

In fact, a recent study by Lam (2025) has suggested that Mandarin finiteness is best understood in terms of semantic dependency, with the so-called non-finite complement being more semantically dependent on the matrix clause than the finite complement. In other words, subject realization, which is arguably a syntactic issue, may require an alternative explanation than finiteness if Mandarin finiteness is indeed more of a semantic than syntactic concept, as proposed by Lam (2025).

In this paper, we leave the finiteness debate aside, and argue that the availability of copy control in Mandarin is best explained by two distinct control mechanisms, one of which involves structure sharing (LFG’s functional control) and the other anaphoric dependency (LFG’s anaphoric control). Control verbs that disallow copy control are functional-control verbs, whereas those that permit copy control are anaphoric-control verbs. This distinction relates to the classical observation by Bresnan (1982a); Mohanan (1983), among others, that the controllee of a functional-control verb cannot be replaced by an overt pronoun, whereas anaphoric-control verbs are more amenable to embedded subject realization. Our explanation for copy control is, therefore, independent of any finiteness claims about Mandarin.

Central to our argumentation is the view that copy control plays an important role in the grammar of Mandarin Chinese by drawing the line of division between functional vs anaphoric control verbs. Control verbs can specify either functional or anaphoric control; therefore, formal control mechanisms must be determined empirically on the basis of the grammatical properties of control verbs. This is in line with the lexicalist view that control and complementation properties are specified in the lexical entries of verbs, as discussed in Section 4.1.3.

In the following sections, we provide a formal analysis of how each control and raising type interacts with the availability of an overt controllee, with respect to the four empirical generalizations in Section 3.4.

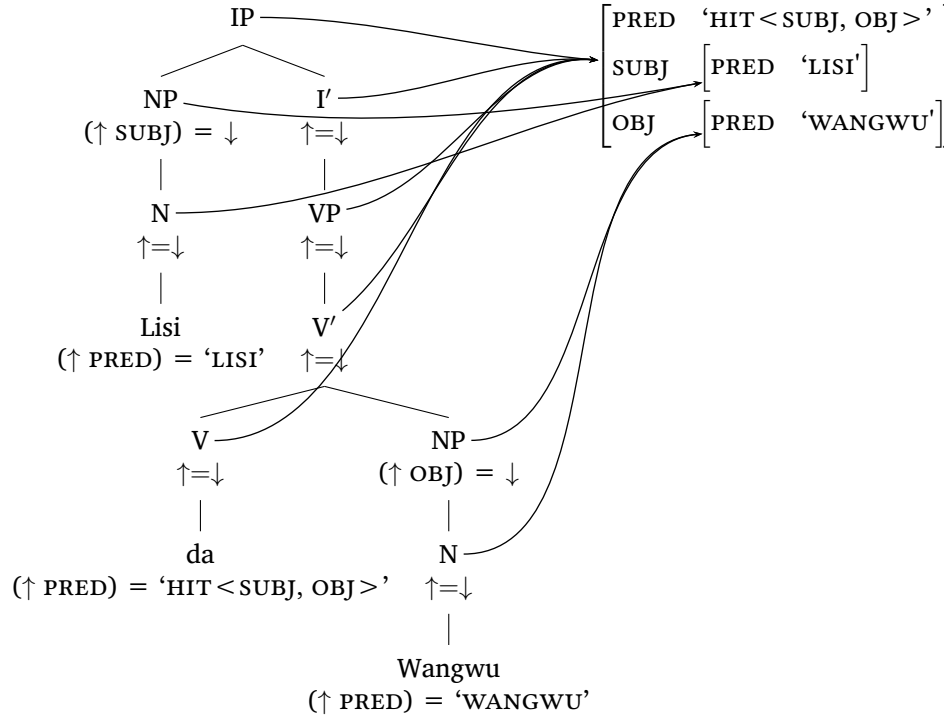
4.1 Introducing LFG

Here we introduce the aspects of the theory of syntax and semantics which will be relevant in our analysis.

4.1.1 LFG’s syntactic structures

Lexical Functional Grammar (LFG: Bresnan 1982b; Bresnan et al. 2016; Dalrymple 2023; Dalrymple et al. 2019) assumes two separate aspects of syntactic structure, represented as separate structures: *constituent structure* (or *c-structure*) is a phrase structure tree representing word order and phrasal groupings, and *functional structure* (or *f-structure*) is an attribute-value structure representing grammatical functions like subject (SUBJ) and object (OBJ). The c-structure tree for the simple Mandarin sentence *Lisi da Wangwu* ‘Lisi hit Wangwu’, annotated with functional equations, is given on the left-hand side of (34), and the f-structure appears on the right-hand side.

(34) C-structure and f-structure for *Lisi da Wangwu* ‘Lisi hit Wangwu’



The down arrow ($\downarrow$) in the functional equations in the c-structure refers to the f-structure of the node on which the annotation appears, and the up arrow ($\uparrow$) refers to the f-structure of its mother node. Thus, the annotation on the NP daughter of the IP node requires the f-structure corresponding to that NP to be the value of the SUBJ attribute in the f-structure for the IP, and the annotation on the I' requires the f-structure for the IP and the f-structure for the I' to be the same. Similarly, the annotation on the N node dominating *Lisi* requires the f-structure for the N and its mother NP node to be the same, and the lexical entry for *Lisi* specifies that the f-structure for the preterminal N which dominates it must have an attribute PRED with value 'LISI'. The f-structure on the right-hand side of (34) satisfies all of the equations in the functional annotations on the c-structure tree. Here we will not be concerned with the many additional f-structure features that would appear in a more complete representation, including nominal features such as PERS (encoding person) and NUM (encoding number), aspect features, and other functional information; our main concern here is with the PRED features and their values, and we will include in our f-structures only those attributes and values which are important in our analysis of copy control in Mandarin.

C- and f-structures are related by a mathematically defined correspondence function ϕ . For example, the NP and N nodes dominating *Wangwu* in (34) correspond, via ϕ , to the OBJ f-structure, as indicated by the curved arrows linking the NP and N nodes to the OBJ f-structure. Similarly, the IP, I', VP, V', and V nodes correspond to the outermost f-structure. The inverse of this function, the inverse correspondence relation ϕ^{-1} , associates an f-structure with its corresponding c-structure nodes: for instance, the OBJ f-structure corresponds via ϕ^{-1} to the NP and N nodes dominated by V'. For a detailed discussion of ϕ and ϕ^{-1} , see Dalrymple et al. (2019: 117, 249).

The value of the PRED feature is special: it is called a *semantic form*. The single quotes surrounding a semantic form indicate uniqueness; semantic forms are instantiated to a unique value for each use, and cannot be multiply specified. Semantic forms also specify argument structure. Arguments which are assigned a semantic role appear in the list of arguments enclosed by angled brackets, and athematic arguments appear outside the angled brackets. In (34), the semantic forms 'LISI' and 'WANGWU' for the names *Lisi* and *Wangwu* do not subcategorize for any arguments and do not include an argument list,

while the semantic form for the transitive verb *da* ‘hit’ specifies that it requires a SUBJ and OBJ; since these arguments appear inside the angled brackets, each is assigned a semantic role.

4.1.2 F-structure well-formedness conditions

F-structures are required to obey the following well-formedness conditions:

(35) Uniqueness Condition

In a given f-structure, every attribute has exactly one value. (Kaplan & Bresnan 1982; see also Asudeh & Toivonen 2015; Dalrymple et al. 2019)

This condition, sometimes called the Consistency Condition, disallows attributes with clashing values: in a particular f-structure, the value of the PERS (person) attribute cannot be simultaneously 1 (first person) and 2 (second person), for example, and the value of the NUM (number) attribute cannot be simultaneously SG and PL. Importantly, the semantic form value of the PRED feature cannot be multiply specified, since semantic forms are uniquely instantiated. This will be important in our analysis of Mandarin verbs disallowing copy control.

(36) Coherence Condition

All governable functions present in an f-structure must occur in the value of a local PRED feature. All functions that have a PRED value must have a θ -role. (Börjars et al. 2019: 22; Crouch et al. 2011; Dalrymple et al. 2019; Kaplan & Bresnan 1982)

Coherence requires that when a semantic form has an argument list, only the required governable functions may appear; for example, the semantic form for an intransitive verb requires SUBJ, and cannot appear with additional governable grammatical functions such as OBJ.

(37) Completeness Condition

All governable functions which are part of the value of a PRED feature must be present in the local f-structure. All functions that have a θ -role must have a PRED value. (Börjars et al. 2019: 21; see also Crouch et al. 2011; Dalrymple et al. 2019; Kaplan & Bresnan 1982)

Completeness requires that when a semantic form has an argument list, all of the required arguments must be present; if a required argument is missing, the f-structure is incomplete. The semantic form for the verb *da* ‘hit’ in (34) requires SUBJ and OBJ, and is incomplete if either of these arguments does not appear.

However, arguments of a predicate are not always supplied by overt phrases appearing in the c-structure tree. To capture the fact that Mandarin is a discourse pro-drop language (Section 2), we propose that the SUBJ of all Mandarin verbs may remain unexpressed in the c-structure, while still being present in the f-structure to satisfy the verb’s subcategorization requirements. (38) is a grammar-wide optional constraint introducing an unpronounced pronominal subject, included in the lexical entries of all Mandarin verbs. This treatment of pro-drop is standard within LFG; for the earliest proposals, see Simpson (1983:126ff.); Bresnan & Mchombo (1987).

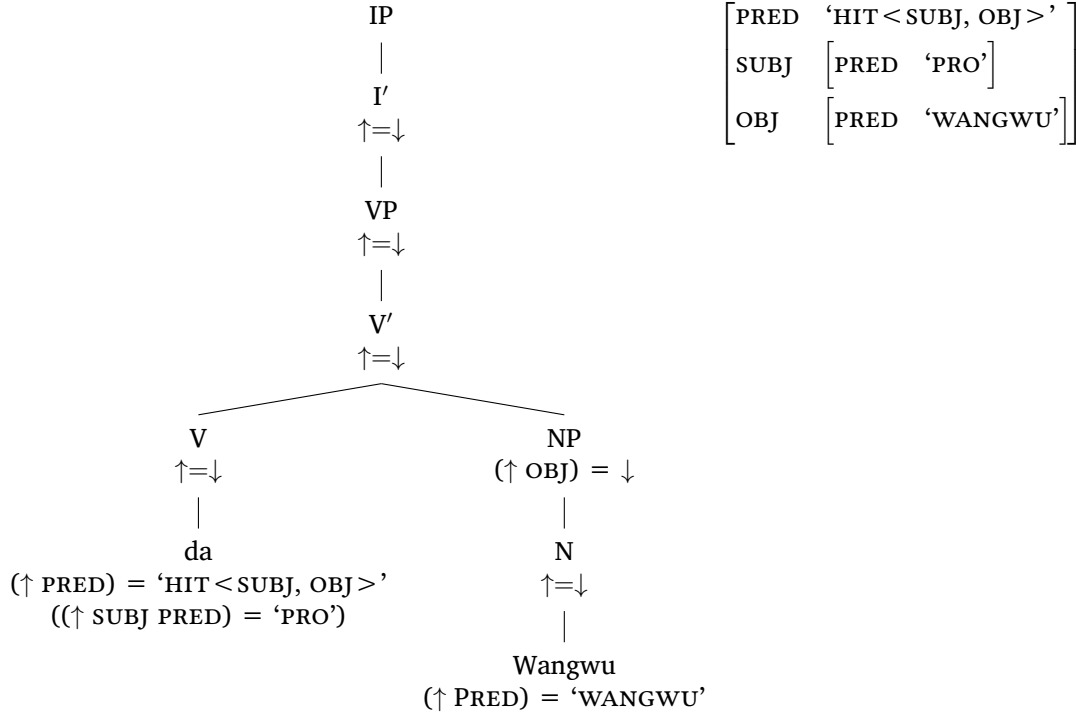
(38) An optional grammar-wide constraint for all Mandarin verbs

$((\uparrow \text{ SUBJ PRED}) = \text{‘PRO’})$

The optionality of the constraint ensures that when the subject of a sentence is realized as, for instance, a full noun phrase, a superfluous PRED ‘PRO’ value for the sentential SUBJ is not introduced, since this

would violate the Uniqueness Condition.¹⁴ Adding the constraint in (38) to the functional description contributed by the verb *da* ‘hit’, we have the c-structure and f-structure in (39) for *da Wangwu* ‘(I/you/he/she/they) hit Wangwu’. Adding the constraint to the verb’s contribution in (34) does not produce any change in the f-structure, since the subject *Lisi* obligatorily contributes PRED ‘LISI’ to the f-structure, and so the optional SUBJ PRED ‘PRO’ contributed by the verb must remain unused.

(39) C-structure and f-structure for *da Wangwu* ‘(I/you/he/she/they) hit Wangwu’



For a detailed introduction to the core concepts of LFG theory, see Belyaev (2023); for an overview of LFG work on Sinitic languages, see Lam et al. (2023).

4.1.3 Functional and anaphoric control in LFG

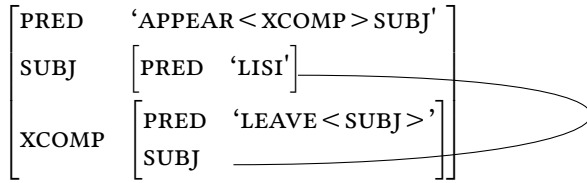
Since Bresnan (1982a), LFG has employed two major approaches to the analysis of control phenomena: functional control and anaphoric control. Vincent (2023) provides an overview of raising and control verbs in an LFG setting.

Functional control LFG analyzes raising verbs as instantiating functional control (Bresnan 1982a; Dalrymple et al. 2019: ch. 15). In functional control, the interpretive dependency between the controller and the controllee is captured through f-structure sharing. The same f-structure plays two different syntactic roles: as the raised argument of the raising predicate, and as the subject of the subordinate predicate. In other words, the matrix verb and the complement verb share an argument.

For example, in the f-structure in (40), the f-structure subject of the English raising verb *appear* is also the subject of the open xCOMP complement, as indicated by the line connecting the f-structure SUBJ of *appear* to the SUBJ of *leave*.

¹⁴A similar constraint could be introduced for Mandarin object pro-drop. However, as noted by Lam (2023: 90–92), object drop in Mandarin is more restricted than subject drop, particularly with respect to the matrix object in complementation constructions. Since object drop is not the focus of this paper, we will not pursue this issue further.

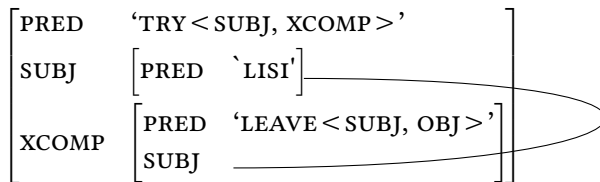
(40) F-structure illustrating functional control for the English raising structure *Lisi appeared to leave*:



The defining characteristic of raising verbs is that they do not assign a θ -role to the raised argument. In the PRED value for *appear*, this is indicated notationally by the placement of the SUBJ argument outside the angled brackets. In (40), the PRED-bearing controller is also the SUBJ within the embedded clause, thereby satisfying the Coherence Condition.

Obligatory control verbs were also analyzed as involving functional control in the earliest LFG analyses (Bresnan 1982a), though as we will see in the next section, some subsequent work treats obligatory control verbs as involving anaphoric control. On a functional control analysis, the f-structure for an English sentence like *Lisi tried to leave* is as in (41), with the control verb *try* subcategorizing for an open XCOMP and sharing a subject with the subordinate verb *leave*:

(41) F-structure illustrating a **functional control** analysis for the English control structure *Lisi tried to leave*:



Here, the same syntactic entity, the f-structure for the subject *Lisi*, is assigned a semantic role in the matrix clause as well as the subordinate clause. This is accounted for straightforwardly in an LFG setting (Asudeh 2005).¹⁵

Anaphoric control In an anaphoric control structure, there is no sharing of syntactic structure or syntactic arguments. Instead, there is an anaphoric relation between the controller and the controllee, and a pronominal represented as the attribute–value pair <PRED, ‘PRO’> appears in the f-structure of the controllee. There is no f-structure sharing, and the PRED-bearing controller does not appear as an argument of the matrix predicate.

Anaphoric control is not viable for raising predicates (Asudeh 2005; Bresnan 1982a; Bresnan et al. 2016; Börjars et al. 2019; Dalrymple et al. 2019; Mohanan 1983). In such settings, the controller/matrix argument of the raising predicate would have no way of receiving a θ -role, and so the Coherence Condition would be violated.¹⁶

Current LFG analyses differ on the analysis of English control verbs (e.g., *try* and *persuade*) as involving functional or anaphoric control. Dalrymple et al. (2019) propose that English control verbs involve anaphoric control rather than functional control: on this view, a verb like *try* subcategorizes for a closed

¹⁵Functional control in LFG has similarities to the treatment of raising verbs via movement in Principles-and-Parameters or Minimalist frameworks (e.g., Adger 2003; Carnie 2021), as well as to the movement theory of control (e.g., Boeckx et al. 2010; Hornstein 1999, 2001; Hornstein & Polinsky 2010). In these movement-based approaches, the matrix and embedded subjects are identical syntactic entities: the matrix subject originates in its θ -position as the subordinate subject and subsequently surfaces in the higher clause through movement. In this setting, the movement theory of control necessitates a re-evaluation of the Theta Criterion to allow an argument to bear more than one thematic role (Hornstein 1999).

¹⁶Anaphoric control is analogous to the PRO theory of control verbs in a Principles-and-Parameters setting, where the controller and the controlled PRO are separate but coindexed syntactic entities. The PRO theory of control is also unsuitable for the analysis of raising predicates, since the controller would be left without a θ -role (see, e.g., Carnie 2021; Chomsky 1981).

COMP and introduces a controlled <PRED, 'PRO'> as the subject of the subordinate clause, as shown in (42).

- (42) F-structure illustrating an **anaphoric control** analysis for the English control structure *Lisi tried to leave*:

$$\left[\begin{array}{l} \text{PRED} \quad \text{'TRY <SUBJ, COMP>'} \\ \text{SUBJ} \quad \left[\begin{array}{l} \text{PRED} \quad \text{'LISI'} \end{array} \right] \\ \text{COMP} \quad \left[\begin{array}{l} \text{PRED} \quad \text{'LEAVE <SUBJ, OBJ>'} \\ \text{SUBJ} \quad \left[\begin{array}{l} \text{PRED} \quad \text{'PRO'} \end{array} \right] \end{array} \right] \end{array} \right]$$

See Asudeh 2005; Bresnan 1982a; Bresnan et al. 2016; Börjars et al. 2019; Mohanan 1983 for functional control analyses of obligatory control verbs; see Dalrymple et al. 2019: ch. 15 for an anaphoric control analysis; see Falk (2001): ch. 5 for a hybrid analysis, where some English control verbs exhibit functional control, and others exhibit anaphoric control.

Functional vs anaphoric control In the cross-linguistic literature, functional and anaphoric control are often distinguished by language-specific properties that follow from the nature of these two different control mechanisms.

Since functional control involves f-structure sharing between the controller and the controllee, any morphosyntactic properties associated with one must also be shared by the other. In Icelandic, for instance, case marking on the matrix subject provides a clear diagnostic for distinguishing functional from anaphoric control verbs (Andrews 1982). In functional control, the matrix subject must bear the same case that the embedded predicate assigns to the subordinate subject, including for 'quirky case' verbs assigning an idiosyncratic case to their subject, because the two subjects share the same syntactic representation. By contrast, when anaphoric control is involved, the matrix subject is not required to obey the case requirements of the embedded predicate, since the matrix and subordinate subjects are syntactically distinct. For Norwegian, Lødrup (2004) argues that one diagnostic for determining whether a Norwegian matrix predicate involves functional or anaphoric control is the impersonal passive construction. Specifically, predicates that can also occur with the impersonal passive—where the sentential subject surfaces as the expletive *det* 'it'—are analyzed as involving anaphoric control rather than functional control. Of course, Lødrup's analysis does not imply that anaphoric control is contingent on the availability of the impersonal passive. Such an assumption would be untenable for languages like Mandarin, which lack this construction altogether. Rather, his investigation supports the broader view that the distinction between functional and anaphoric control must be determined empirically, based on the grammatical properties of individual matrix predicates in each language.

Another diagnostic property distinguishing the two types of control is whether the controllee can be replaced by an overt pronoun or noun phrase (e.g., Bresnan 1982a; Börjars et al. 2019; Mohanan 1983). For instance, Mohanan (1983) argues that English and Malayalam participial adjunct clauses instantiate functional control, since the controllee cannot be expressed overtly. Börjars et al. (2019: 119) contend that the English verb *want* exemplifies anaphoric control, in contrast to functional-control verbs such as *try*. They argue that one piece of evidence for this distinction comes from sentences like *Oscar wants Fred to buy olives*, where *Fred* serves as the overt subordinate subject—which, unlike ordinary objects in English, cannot undergo passivization—alongside sentences like *Oscar wants to buy olives*, where the controllee is unexpressed. As we will show, the possibility for an overt controllee in Mandarin plays a crucial role in distinguishing between functional and anaphoric control.

Semantics of anaphoric control Recent work (e.g., Dalrymple et al. 2019; Haug 2013, 2014) distinguishes *obligatory anaphoric control* from *quasi-obligatory anaphoric control*. Obligatory anaphoric control

models exhaustive control, and quasi-obligatory anaphoric control models partial control (Haug 2013). These proposals make use of a version of Discourse Representation Theory, *Partial Compositional Discourse Representation Theory* (PCDRT: Haug 2014), to model coreference relations. In this paper, we abstract away from most of the formal technical details of PCDRT and the theory of anaphoric reference, keeping the use of formal tools to the minimum necessary for the discussion. However, two formal details are directly relevant to our forthcoming analysis: (i) discourse referents are associated with a syntactically accessible object which we refer to as an INDEX; (ii) the anaphoric resolution function $\mathcal{R}$ maps the INDEX of an anaphor to a pair consisting of an antecedent INDEX and a semantic relation. Following Haug (2013), we represent this as $\mathcal{R}(\text{anaphoric_index}) \mapsto \langle \text{antecedent_index}, \text{semantic_relation} \rangle$.¹⁷ For exhaustive anaphoric control verbs (for instance, in the lexical entry of the English verb *try*) there is an anaphoric resolution constraint $\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{ INDEX}) \mapsto \langle ((\uparrow \text{SUBJ})_\sigma \text{ INDEX}), \lambda x. \lambda y. x = y \rangle$, where the sigma symbol σ is a projection function from a syntactic f-structure to its corresponding semantic s-structure, as discussed in Dalrymple et al. (2019). Here, the constraint specifies that the controllee COMP SUBJ is an anaphor whose antecedent is the controller SUBJ, and that the semantic relation between the two, captured by $\lambda x. \lambda y. x = y$, is strict coreference. For partial control, the lexical entry of *want* includes the anaphoric resolution constraint $\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{ INDEX}) \mapsto \langle ((\uparrow \text{SUBJ})_\sigma \text{ INDEX}), \mathcal{B} \rangle$.¹⁸ Following Haug (2013), we assume that partial control involves a bridging inference $\mathcal{B}$ (Nouwen 2003), which resolves an anaphor to an entity that is related to—but not necessarily identical with—a salient discourse referent. For instance, in the sentence *The chair wants to meet at 7 pm*, the partial control verb *want* triggers an interpretation in which *the chair* primes a plural referent, namely the chair together with the committee that the chair heads. This inclusion relation is supplied by a bridging relation such as *chair-of*, made salient by the discourse. In this case, the bridging inference $\mathcal{B}$ is instantiated as $\lambda y. \lambda x. \text{chair-of}(x, y)$ by the context. (43) summarizes the definition of the $\mathcal{R}$ function and its applications in EC and PC situations.¹⁹

- (43) a. General definition of $\mathcal{R}$
 $\mathcal{R}(\text{anaphoric_index}) \mapsto \langle \text{antecedent_index}, \text{semantic_relation} \rangle$
- b. $\mathcal{R}$ for exhaustive subject-control verbs
 $\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{ INDEX}) \mapsto \langle ((\uparrow \text{SUBJ})_\sigma \text{ INDEX}), \lambda x. \lambda y. x = y \rangle$
- c. $\mathcal{R}$ for partial subject-control verbs
 $\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{ INDEX}) \mapsto \langle ((\uparrow \text{SUBJ})_\sigma \text{ INDEX}), \mathcal{B} \rangle$, where the bridging inference $\mathcal{B}$ is instantiated by context.

4.2 Mandarin raising constructions: Functional control

Empirical Generalization Three states that raising constructions do not allow overt controllees, regardless of the presence (or absence) of an intervening adverbial. This is exemplified in (44), where the raising verb is *haoxiang* ‘appear’, one of the raising verbs used in our Experiment 1.

¹⁷We follow Haug (2013) in defining $\mathcal{R}$. Dalrymple et al. (2018) provide a slightly different definition of $\mathcal{R}$ and a discussion of partial control which is similar in spirit but notationally different from Haug’s analysis.

¹⁸Here we omit discussion of the logophoric component of partial and exhaustive control. Haug (2013) provides a full analysis which integrates a logophoric-binding constraint, treating the controllee as a logophor and its controller as a logocenter, thereby ensuring that the controllee identifies its controller within the matrix clause. Since we are not concerned with the logophoric aspect of anaphoric binding in control, we simplify the formal treatment here by defining $\mathcal{R}$ for PC solely in terms of binding constraints on the controller and controllee.

¹⁹Asudeh (2005) discusses another way of analyzing PC via functional control, where partial coreference is modeled in the semantics. Since our paper is less concerned with how PC can also be handled via functional control, we will not explore this alternative further here.

- (44) lisi haoxiang (zai huiyi qijiang) (*ta) jiandao wangwu
 Lisi appear at meeting during 3SG see Wangwu
 ‘Lisi appeared to have seen Wangwu during the meeting.’

In (44), although the controller *Lisi* has a PRED value, its local predicate *haoxiang* ‘appear’ is a raising verb and does not assign a θ -role to it.

Following standard LFG assumptions about raising verbs requiring functional control, we propose that (44) is associated with the (simplified) f-structure in (45), where we have omitted the details of the representation of the adverbial *zai huiyi qijiang* ‘during the meeting’, along with any other information not relevant to the issues at hand.

- (45)

[	PRED	‘APPEAR < XCOMP > SUBJ’	]
	SUBJ	[PRED ‘LISI’]	
		[PRED ‘SEE < SUBJ, OBJ >’]	
	XCOMP	SUBJ	
		OBJ [PRED ‘WANGWU’]	
	ADJ	{ [...] }	
-

The lexical entry for *haoxiang* ‘appear’ is displayed in (46). The SUBJ function appears outside the angled brackets in the argument list of the semantic form, signifying that SUBJ is not a semantic argument of the predicate. The second line of the lexical entry is a functional-control equation establishing a structure-sharing relationship between the f-structure of the matrix SUBJ and that of the embedded SUBJ, with the matrix SUBJ being the (athematic) controller.

- (46) Lexical entry of *haoxiang* ‘appear’:
haoxiang ‘appear’ V (↑ PRED) = ‘APPEAR < XCOMP > SUBJ’
 (↑ SUBJ) = (↑ XCOMP SUBJ)

The structure-sharing constraint (↑ SUBJ) = (↑ XCOMP SUBJ) rules out the possibility of an overt embedded SUBJ (controllee); this is because an overt noun phrase or pronoun as the SUBJ of the complement clause would contribute its own PRED, causing a violation of the Uniqueness Condition (35). The ill-formed f-structure in (47), which contains a pronominal embedded subject alongside structure sharing, illustrates such a violation. Due to structure sharing, the values of the matrix SUBJ and embedded SUBJ attributes each end up containing both <PRED, ‘LISI’> and <PRED, ‘PRO’>, creating a clash. The possibility of an overt controllee is thus ruled out by the mechanism of functional control.

- (47) Ill-formed f-structure/Uniqueness Condition violation: functional control with overt controllee
 **Lisi haoxiang...ta jiandiao wangwu* ‘Lisi appeared...he to have seen Wangwu’

- *

[	PRED	‘APPEAR < XCOMP > SUBJ’	]
	SUBJ	[PRED ‘LISI’]	
		[PRED ‘SEE < SUBJ, OBJ >’]	
	XCOMP	SUBJ [PRED ‘PRO’]	
		OBJ [PRED ‘WANGWU’]	
	ADJ	{ [...] }	
-

4.3 Mandarin transitive subject control constructions: Functional control

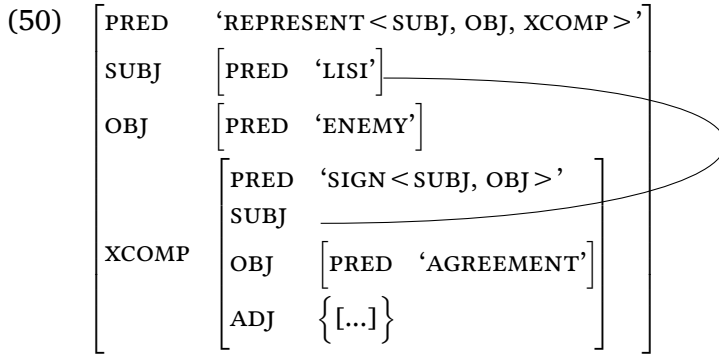
Empirical Generalization Four in (33) states that Mandarin transitive subject-control verbs, such as *daibiao* ‘represent’ and *dayin* ‘promise’, cannot license copy control (regardless of their EC/PC properties). We analyse these as functional-control verbs, where f-structure sharing rules out the possibility of an overt controllee: as with raising verbs, an overt controllee would contribute a superfluous PRED and thus violate the Uniqueness Condition, in the same way as (47).²⁰

- (48) lisi daibiao difang [zai zuihou qixian daolai zhiqian] (*ta) qianshu xieyi
 Lisi represent enemy.PL at final deadline arrive before 3SG sign agreement
 Lit: ‘Lisi represented the enemy to sign the agreement before the final deadline.’
 ‘Lisi signed the agreement on behalf of the enemy before the final deadline.’

- (49) Lexical entry of *daibiao* ‘represent’:

daibiao ‘represent’ V (↑ PRED) = ‘REPRESENT < SUBJ, OBJ, XCOMP >’
 (↑ SUBJ) = (↑ XCOMP SUBJ)

SUBJ, OBJ and XCOMP are arguments of *daibiao* ‘represent’. The lexical entry contains a subject-control equation (↑ SUBJ) = (↑ XCOMP SUBJ), specifying f-structure sharing between the SUBJ of *daibiao* and the SUBJ of the XCOMP. This lexical entry gives rise to the f-structure in (50) for (48); we have again simplified by omitting the details of the adverbial.



To sum up, we claim that the inability for raising and transitive subject-control verbs to appear with overt subordinate subjects is explained on the basis that both require functional control.

4.4 Mandarin exhaustive control constructions: Anaphoric control

In this section, we focus on the behavior of EC verbs, as summarized by Empirical Generalization One in (30). We have seen that EC verbs, such as *xiangbanfa* ‘try’ and *deyi* ‘manage’, allow copy control if there is an intervening adverbial. We propose that EC verbs exemplify obligatory anaphoric control.

- (51) lisi deyi [zai bi-men huiyi jinxing qijiang] (ta) jiandao-le wangwu
 Lisi manage at closed-door meeting on.going during 3SG see-PRF Wangwu
 ‘Lisi managed to meet Wangwu during the closed-door meeting.’

We first present the formal analysis for EC verbs, using *deyi* ‘manage’ as an example. As discussed in Section 4.1.3, anaphoric control involves an anaphoric relation between a controller and its controllee.

²⁰We follow Lam (2022) in analyzing Mandarin control verbs, including EC verbs, as bi-clausal, in contrast to Grano’s (2015) mono-clausal analysis (see also Huang 2018).

The controllee subject is represented as a pronoun which is present in the f-structure but may be absent in the c-structure. Since EC verbs involve anaphoric control, their lexical entries include the anaphoric resolution function $\mathcal{R}$ in (43).

The f-structure for example (51) is shown in (52). (53) is the proposed lexical entry for *deyi* ‘manage’. For ease of reference, we have numbered the constraints.

$$(52) \left[\begin{array}{ll} \text{PRED} & \text{'MANAGE < SUBJ, COMP >'} \\ \text{SUBJ} & \left[\begin{array}{l} \text{PRED} \quad \text{'LISI'} \end{array} \right] \\ \text{COMP} & \left[\begin{array}{l} \left[\begin{array}{l} \text{PRED} \quad \text{'SEE < SUBJ, OBJ >'} \\ \text{SUBJ} \quad \left[\begin{array}{l} \text{PRED} \quad \text{'PRO'} \end{array} \right] \\ \text{OBJ} \quad \left[\begin{array}{l} \text{PRED} \quad \text{'WANGWU'} \end{array} \right] \end{array} \right] \\ \text{ADJ} \quad \{ \dots \} \end{array} \right] \end{array} \right]$$

(53) Lexical entry of *deyi* ‘manage’:

deyi ‘manage’ V

1st constraint $(\uparrow \text{PRED}) = \text{'MANAGE < SUBJ, COMP >'}$

2nd constraint $\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{ INDEX}) \mapsto \langle ((\uparrow \text{SUBJ})_\sigma \text{ INDEX}), \lambda x. \lambda y. x = y \rangle$

3rd constraint $\text{REALIZE}(\uparrow \text{COMP SUBJ}) \Rightarrow [(\uparrow \text{COMP ADJ} \in) <_f (\uparrow \text{COMP SUBJ})]$

The first constraint states that *deyi* ‘manage’ subcategorizes for SUBJ and COMP. The second constraint uses the $\mathcal{R}$ function, as discussed in Section 4.1.3, to establish an antecedency relationship between the controllee and controller, identifying the controllee’s antecedent as the matrix SUBJ (*Lisi* in (51)); the specification of the semantic relation as $\lambda x. \lambda y. x = y$ ensures complete coreference between the controller and the controllee. The $\mathcal{R}$ constraint holds irrespective of whether the controllee is overt: whether the controllee is a null pronoun contributed by the subordinate verb (as specified by the constraint in (38)) or an overt pronoun, it must be coreferential with the controller SUBJ.

The third constraint is an implicational constraint (with $\Rightarrow$ as the implicational operator) that models the licensing condition for the intervening adverbial. An implicational constraint consists of two parts, the condition and the consequent, related by the operator $\Rightarrow$. The condition $\text{REALIZE}(\uparrow \text{COMP SUBJ})$ uses the function REALIZE , defined by Asudeh (2009: 111):

(54) $\text{REALIZE}(f)$ holds if and only if $\phi^{-1}(f) \neq \emptyset$.

The definition adopts the inverse relation $\phi^{-1}(f)$, which associates f-structures with their corresponding c-structural nodes, as discussed in Section (4.1). The constraint holds of an f-structure f if and only if there is some c-structural node that corresponds to f ; for more details of how ϕ^{-1} works, see Dalrymple et al. (2019: 247–250). As such, $\text{REALIZE}(\uparrow \text{COMP SUBJ})$ holds when the embedded subject is overtly realized in the c-structure—in other words, in copy control.

The consequent $[(\uparrow \text{COMP ADJ} \in) <_f (\uparrow \text{COMP SUBJ})]$ involves a relation known as *f-precedence* $<_f$, as defined by Kaplan & Zaenen (1989).²¹

(55) F-precedence (Kaplan & Zaenen 1989; cited from Dalrymple et al.: 257)

f f-precedes g ($f <_f g$) if and only if for all $n_1 \in \phi^{-1}(f)$ and for all $n_2 \in \phi^{-1}(g)$, n_1 c-precedes n_2 .

An f-structure f f-precedes an f-structure g if and only if all of the nodes corresponding to f c-precede all of the nodes corresponding to g . This definition also appeals to the inverse relation ϕ^{-1} to relate an f-structure to a c-structure, and builds on the notion of c-precedence in (56):

²¹Bresnan et al. (2016: 213) provide an alternative definition of f-precedence which could also be used here.

(56) C-precedence (Dalrymple et al. 2019: 256)

A c-structure node n_1 c-precedes a node n_2 if and only if n_1 does not dominate n_2 , n_2 does not dominate n_1 , and all nodes that n_1 dominates precede all nodes that n_2 dominates.

Taken together, these definitions allow us to interpret the consequent $[(\uparrow \text{COMP ADJ } \epsilon) <_f (\uparrow \text{COMP SUBJ})]$ as stating that there is an adverbial within the complement clause and that this adverbial f-precedes the embedded subject.²²

Using the inverse relation ϕ^{-1} to constrain c-structural relations via f-structural constraints is common in LFG (e.g., Asudeh 2009). One motivation for adopting this formal strategy here lies in our view of control as a lexically determined phenomenon, consistent with the general LFG approach (Section 4.1.3). Because control constraints are specified in the lexicon, and because the lexicon directly provides information for building the f-structure (Section 4.1), any further lexically sanctioned constraints on c-structural relations are most naturally mediated by the f-structure. Combining the condition and the consequent, the implicational constraint in the lexical entry of *deyi* ‘manage’ imposes the requirement that whenever the embedded subject is overtly realized, an intervening adverbial must precede it. This precisely implements Empirical Generalization One (for EC verbs) within LFG’s formal architecture.

4.5 Mandarin partial control constructions: Anaphoric control

According to Empirical Generalizations One and Two, for constructions involving PC verbs (e.g., *zhunbei* ‘prepare’ and *dasuan* ‘intend’), when the controller and controllee share identical reference, an intervening adverbial is required to license copy control; however, copy control is possible without an intervening adverbial when the controller is a subset of entities denoted by the controllee. (57) illustrates these conditions.

- (57) a. lisi zhunbei [zai bi-men huiyi jinxing qijian] (ta) huijian wangwu
 Lisi prepare at closed-door meeting on.going during 3SG meet Wangwu
 ‘Lisi prepared to meet Wangwu during the closed-door meeting.’
 b. lisi zhunbei (tamen) yiqi huijian wangwu
 Lisi prepare 3PL together meet Wangwu
 ‘Lisi prepared to meet Wangwu together.’
 c. lisi zhunbei (*ta) huijian wangwu
 Lisi prepare 3SG meet Wangwu
 ‘Lisi prepared to meet Wangwu.’

(58) presents the lexical entry for the PC verb *zhunbei* ‘prepare’, and (59) shows the simplified f-structure for (57a), instantiating anaphoric control. According to the first constraint in (58), *zhunbei* subcategorizes for SUBJ and COMP. The second constraint makes use of the anaphoric resolution function $\mathcal{R}$, which maps an anaphoric index to a pair consisting of an antecedent index and a coreference constraint, as discussed in Section 4.1.3. Here, the coreference constraint is the bridging-inference relation $\mathcal{B}$. In the PC reading of (57b), $\mathcal{B}$ may be instantiated as $\lambda y. \lambda x. \text{member-of}(x, y)$, if the context supports an interpretation in which *Lisi* was a member of a group that met *Wangwu*. On the other hand, $\mathcal{B}$ can also be instantiated as complete coreference, as in (57a), where the pronoun *ta* refers to *Lisi* alone.

²²Following the word-order requirements of Mandarin, we assume that an adverbial phrase must appear within the clause which it modifies: in other words, non-local or long-distance adverbial modification is not possible. Thus, if an adverbial phrase modifies a clause and precedes the subject of that clause, it appears immediately before the subject.

(58) Lexical entry of *zhunbei* ‘prepare’:

	<i>zhunbei</i> ‘prepare’ V
1st constraint	$(\uparrow \text{PRED}) = \text{‘PREPARE <SUBJ, COMP>’}$
2nd constraint	$\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{INDEX}) \mapsto \langle (\uparrow \text{SUBJ})_\sigma \text{INDEX}, \mathcal{B} \rangle$
3rd constraint	$[\text{REALIZE}(\uparrow \text{COMP SUBJ})] \ \& \ [\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{INDEX}) \mapsto \langle (\uparrow \text{SUBJ})_\sigma \text{INDEX}, \lambda x.\lambda y.x = y \rangle]$ $\Rightarrow [(\uparrow \text{COMP ADJ}) \in] <_f (\uparrow \text{COMP SUBJ})]$

(59)

PRED	‘PREPARE <SUBJ, COMP>’
SUBJ	[PRED ‘LISI’]
COMP	[PRED ‘MEET <SUBJ, OBJ>’]
	SUBJ [PRED ‘PRO’]
	OBJ [PRED ‘WANGWU’]
	ADJ { [...] }

The third constraint in (58) models the condition on the intervening adverbial. This implicational constraint consists of a condition (antecedent) and a consequent. The condition here contains two conjuncts: if both conjuncts are satisfied, the consequent must hold. In this case, the second conjunct $[\mathcal{R}((\uparrow \text{COMP SUBJ})_\sigma \text{INDEX}) \mapsto \langle (\uparrow \text{SUBJ})_\sigma \text{INDEX}, \lambda x.\lambda y.x = y \rangle]$ plays a crucial role in regulating the behavior of PC verbs: only when the embedded subject is overtly realized and stands in complete coreference with the matrix subject does the constraint require the presence of an intervening adverbial in the complement clause preceding the embedded subject. This formal setup yields the following consequences: (i) In (57a) and (57c), where the embedded subject is overt and exhaustively corefers with the matrix subject, an intervening adverbial must be present for the sentence to be grammatical; (ii) When the embedded subject is overt but does not exhaustively corefer with the matrix subject, an intervening adverbial is not required, as illustrated in (57b). Taken together, the constraint precisely captures Empirical Generalization One (for PC verbs) and Empirical Generalization Two within LFG’s formal architecture.

To sum up, we have presented a formal analysis grounded in anaphoric control to model the copy-control behavior of both EC and PC verbs. This analysis accounts for the different scenarios captured by our empirical generalizations. Crucially, EC and PC verbs impose implicational constraints involving the presence of an adverbial phrase which yield the correct empirical predictions. The fact that very similar constraints apply across verb types reduces theoretical complexity and maintains the elegance of the analysis.

5 Implications and future research

5.1 Copy control vs processing-based resumption

In our formal analysis, the requirement for an intervening adverbial for EC and PC verbs, represented by implicational constraints, is integrated as part of the formal grammar of Mandarin. As discussed in Section 2.2, Grano (2015: 147) regards overt controllees as resumptive pronouns whose optionality is conditioned by the presence vs absence of an intervening adverbial, consistent with the finding that the acceptability of English resumptive pronouns improves with increased syntactic distance between the site of resumption and the landing site. In fact, English resumptive/intrusive pronouns have been analyzed independently by Asudeh (2012) as a purely processing-based phenomenon that is not licensed by the grammar of the language but arises from incremental production. Thus, any claim that Mandarin

overt controllees resemble English resumptives implies that Mandarin copy control may also be purely processing-based and thus may not be grammatically licensed.

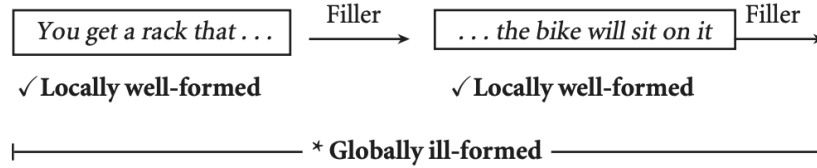


Figure 3: Incremental processing of English filler–gap dependency involving a resumptive (Asudeh 2012: 293)

According to Asudeh (2012: ch. 11), leveraging the fact that LFG can model fragments (Bresnan et al. 2016), an utterance is incrementally produced in fragments (“chunks”), where each new chunk must be locally grammatical, leading to incremental generation of a full utterance that satisfies local grammaticality at every incremental step.²³ Underpinning Asudeh’s incremental theory are the differences between “local grammaticality” and “global grammaticality”. The production of an English utterance involving a resumptive pronoun in filler–gap dependencies such as relative-clause constructions (for example, *You get a rack that the bike will sit on it*) involves the incremental processes shown in Figure 3. The first chunk, *You get a rack that...*, contains the filler, which is the head noun *a rack*; in the upcoming incremental step, the filler is supposed to be integrated into the overall structure of the utterance by filling in a gap in the relative clause. The first chunk is considered locally well-formed with the predicate-argument relations required by the main verb *get* being satisfied; as the production system continues to generate the second chunk with the relative clause, the gap after the preposition *on* is filled by the resumptive pronoun *it*. The second chunk is still locally well-formed, since the predicate-argument relations required by *sit on* are fulfilled; however, due to the insertion of the resumptive pronoun, the filler in the first chunk can no longer be integrated into the overall structure. The grammar ultimately fails to license the overall structure of the utterance, which is therefore globally ill-formed—consistent with the judgments of English native speakers reported by Asudeh, who draws on previous acceptability studies of English speakers’ responses to resumptive pronouns in filler–gap dependencies. Asudeh identifies the distance between the filler and the gap as one of the factors motivating English resumptives in incremental processing, consistent with Grano’s observation that the acceptability of English resumptives increases with greater distance between the two sites.

However, there are substantial differences between Mandarin copy control and English resumptive strategies. First, while previous studies indicate that English filler–gap dependencies with resumptive pronouns are judged by native speakers to be globally ill-formed (Asudeh 2012), Mandarin copy control involving EC and PC verbs are judged as well-formed, as demonstrated by our experimental findings in Section 3. This suggests that Mandarin copy control is licensed by the grammar of the language. Furthermore, a processing-based account alone cannot explain why Mandarin copy control varies across control and raising types, which represent grammatically licensed distinctions. Given an intervening adverbial of the same length, a purely processing-based account would incorrectly predict that transitive subject-control verbs favor copy control more strongly than intransitive EC and PC verbs, because of the greater distance between the controller and controllee. We therefore maintain that the treatment of English resumptives should not be extended to Mandarin copy control, and that the constraints governing Mandarin copy control must be integrated into the grammar of the language rather than delegated to incremental processing mechanisms.

²³Similar incremental processes are also posited for parsing, which involves a few more independently motivated assumptions. For details, see Asudeh (2012: 301).

5.2 Revised obviation as motivating constraints on copy control

Rather than construing Mandarin overt controllees as resumptives, Hu et al.’s (2001) obviation principle, discussed in Section 2.2, may be reconsidered as a motivation for the grammaticalization of the requirement for the presence of an intervening adverbial in copy control. According to Hu et al.’s (2001) obviation principle, an overt pronoun tends to obviate from its closest prominent noun phrase. Consequently, when an intervening adverbial containing a noun phrase appears between an overt controllee and its controller, the controller is no longer the closest prominent noun phrase to the pronoun, thereby satisfying the obviation principle. As mentioned in Section 2.2, this principle has been criticized for yielding incorrect predictions in non-control sentences. Nevertheless, it may be worth exploring whether a reformulated version of the obviation principle—one that limits its obviative tendency to control environments—could offer a motivation for the syntactic constraints that we have proposed. In order for copy control with exhaustive coreference to be possible, an intervening adverbial is required to attenuate the obviation effects. By contrast, in cases with partial coreference, where the controller and overt controllee are not fully coreferential, the obviation principle does not apply, and an intervening adverbial is not required.²⁴

5.3 Interspeaker variation and our lexicalist account

We have established four empirical generalizations on the basis of controlled experiments, and we have provided a formal analysis of copy control across various control and raising types. While our findings broaden the empirical landscape of Mandarin copy control, they also diverge in places from patterns reported in some earlier studies (Section 2). Given our lexicalist account, we may view these divergences as interspeaker variation.

We have proposed that Mandarin copy control is modeled by two broad mechanisms: functional control and anaphoric control. Anaphoric control permits copy control, whereas functional control disallows it. On our account, the availability of copy control rests on the lexical specification of the matrix verb. Under this view, interspeaker differences can be modeled as idiolectal differences in lexical entries—a well-attested phenomenon. Individual native speakers differ in the size and content of their mental lexicons, as shown by work on the mental lexicon and vocabulary knowledge (e.g., substantial between-speaker variation in native vocabulary size and composition, and robust word-prevalence effects showing that the proportion of speakers who know a given word varies widely: Aitchison 2012; Brysbaert et al. 2019, 2016). Relatedly, research on individual differences in lexical quality and native grammatical attainment demonstrates that adult (monolingual) speakers need not share identical lexical/grammatical representations (Andrews 2015; Dąbrowska 2012). On this approach, one speaker may represent a given EC verb with an anaphoric-control entry (thereby allowing copy control as per Grano 2015; Hu et al. 2001), whereas another may store the same verb with a functional-control entry (thereby blocking copy control as per He 2020; Huang 2022, 2024; Paul 2018). Similarly, one speaker’s PC verb entry may contain an implicational constraint specifying the conditions for the intervening adverbials (as per Grano 2015), while another speaker may not have this constraint, thus allowing copy control without an intervening adverbial (as per Li 2024; Zhang 2016). Therefore, our lexicalist approach to copy control can yield possible interspeaker differences across different studies without altering the core architecture of our theory, namely the distinction between functional vs. anaphoric control.

²⁴Such an explanation may not extend to Telugu, discussed in Section 2.5, which involves copy control in adjunct control rather than complement control as in Mandarin. Further, the examples of Telugu copy control provided by Haddad (2009) always involve another noun-phrase argument between the controller and overt controllee, so it is hard to tell whether a revised version of Hu et al.’s (2001) obviation principle would apply to Telugu.

5.4 Hybrid model of control in LFG and the two-tiered theory in Minimalism

In Section 1, we noted that the Principles-and-Parameters and Minimalist traditions distinguish two major approaches to control: the PRO-based theory and the movement-based theory of control. For purposes of cross-framework comparison, we may draw parallels between PRO-based analyses and LFG’s anaphoric control on the one hand, and between movement-based analyses and LFG’s functional control on the other (Vincent 2023). While PRO-based and movement-based theories are typically presented as competing treatments of obligatory control (Hornstein & Polinsky 2010; Landau 2024a), we are not aware of any work arguing that a single language, within the domain of complement control, requires a movement-based analysis for one class of control verbs and a PRO-based analysis for another.²⁵ The situation is different in LFG where, since the inception of the theory, functional control and anaphoric control have been understood as complementary mechanisms operating within a unified theoretical system. Together, they capture the diversity of control patterns both cross-linguistically and within individual languages.

Some existing studies (e.g., Li 2024; Zhang 2016) have appealed to Landau’s (2015; 2024a) two-tiered theory of control (TTC)—a PRO-based theory—to analyze Mandarin copy control. TTC offers a dual system framed around the classical property vs proposition distinction, namely whether the clausal complement of a control verb denotes a property or a proposition (see, e.g., Chierchia 1985; Higginbotham 1983). The theory distinguishes predicative control and logophoric control. Under predicative control, the complement denotes a property predicated of the controller, and Landau argues that all EC verbs fall into this class (Landau 2024a: 36). Under logophoric control, the complement denotes a proposition, and the controller serves as the antecedent for a logophoric element in the embedded clause; on Landau’s view, all PC verbs instantiate this type (Landau 2024a: 37).

In contrast to TTC, LFG’s functional control and anaphoric control are formally compatible with both property-based and propositional treatments of clausal complements (see Asudeh 2005 for a property-based analysis and Dalrymple et al. 2019 for a propositional one). TTC predicts that only PC verbs (under logophoric control) should allow the embedded subject to be lexicalized, whereas EC verbs (under predicative control) should not (Landau 2024a: 44–46). As our findings show, however, this prediction is not borne out for Mandarin copy control: both EC and PC verbs permit lexicalized embedded subjects, albeit under different structural and interpretive conditions. Our LFG analysis—drawing on the functional vs anaphoric control distinction and employing an implicational constraint—successfully models these patterns and captures the empirical behavior of Mandarin control verbs across classes.

5.5 Toward a typology of anaphoric control for copy control

Our analysis of Mandarin copy control has implications for the typology of anaphoric control. As discussed in Section 4.1.3, Dalrymple et al. (2019) employ anaphoric control to analyze English control verbs such as *try*. They propose the lexical entry in (60) for *try* (Dalrymple et al. 2019: 567):

(60) Lexical entry of English control verb *try* (anaphoric control analysis):

try V (↑ PRED) = ‘TRY < SUBJ, COMP >’
(↑ COMP SUBJ PRED) = ‘PRO’

As shown in (60), the lexical entry includes a constraint that obligatorily assigns a pronominal PRED value to the SUBJ of the complement clause COMP. A key consequence of this specification is that the complement clause cannot contain an overt subject, such as an overt pronoun (*he/she/it/they*) or a full NP. Any overt subject would introduce a superfluous PRED value for the complement SUBJ, thereby violating the requirement discussed in Section 4.1 that semantic forms must be instantiated uniquely:

²⁵Movement-based approaches may, however, recognize a role for PRO in non-obligatory control (Hornstein & Polinsky 2010).

that is, a PRED value cannot be multiply specified, even if those multiple specifications happen to take the same form PRO. Thus, the constraint ($\uparrow$ COMP SUBJ PRED) = ‘PRO’ for English anaphoric control verbs effectively rules out the possibility of copy control in English.

By contrast, Mandarin anaphoric-control verbs do not include this constraint. Instead, Mandarin—being a pro-drop language—employs the optional constraint (($\uparrow$ SUBJ PRED) = ‘PRO’) for all its verbs, as discussed in Section 4.1.2. This setup, together with the constraints formalized in Section 4, allows both pro-drop and copy control to be grammatically licensed in Mandarin. This contrast has broader typological implications. Similar formal strategies may extend to other pro-drop languages, such as Hungarian (É. Kiss 2012), that permit copy control (Section 1). For non-pro-drop languages that allow copy control, such as Zapotec (Black 1996) as exemplified in Section 1, it is possible to introduce an optional pronominal constraint to the lexical entry of an anaphoric-control verb, as shown in (61):

(61) (($\uparrow$ COMP SUBJ PRED) = ‘PRO’)

In copy-control contexts, where the embedded subject is overt and specifies its own PRED value, this optional constraint remains inactive. However, in non-copy-control contexts, where the embedded subject is null, this constraint supplies a pronominal PRED value, satisfying any subject subcategorization requirements of the embedded verb and thus fulfilling the Completeness Condition (37).

Regarding the possible forms of the overt controllees (e.g., personal pronoun, complemented pronoun, logophoric pronoun, full noun phrase), as surveyed in Section 1, we expect that one of the determining factors comes from the distributions of noun phrases and various kinds of pronouns in the relevant languages, subject to their respective binding conditions. For example, full noun phrases in Mandarin are known to follow Principle C (Huang et al. 2009: 329), which requires them to be free from c-commanding expressions.²⁶ This restriction explains why Mandarin overt controllees do not appear as full noun phrases, since binding by a c-commanding matrix subject would violate Principle C. An apparent exception is San Lucas Quiaviní Zapotec, where a full noun phrase can serve as an overt controllee in complement control, as illustrated in (2). By contrast, complemented pronouns (comprising a pronoun and a noun complement) are known not to obey Principle C (Szabolcsi 2009a: 16), which may account for their availability as overt controllees in languages such as Hungarian. We anticipate that future research will further elucidate the typology of anaphoric control and its relation to copy control, as well as cross-linguistic variation in the forms of overt controllees.

6 Conclusions

This paper has examined Mandarin copy control and its interaction with other control-related properties in the language. We have shown that the empirical landscape of Mandarin copy control is considerably more intricate than previously recognized. Our findings are distilled into four empirical generalizations, each supported by formal acceptability-judgment experiments. Our study departs from previous approaches, nearly all of which appeal to finiteness-based explanations—often relying on divergent assumptions about whether Mandarin encodes finiteness and, if so, how finiteness should be characterized. This move also aligns with broader cross-linguistic developments in the theory of control: given the growing body of work on finite control across languages, there is increasing consensus that (non-)finiteness is no longer a necessary or defining property of control relations (Landau 2024a: 22). Instead, we adopt two well-established control mechanisms in LFG—functional control and anaphoric control—to account for the observed patterns, linking the availability of overt controllees with other properties, including exhaustive control, partial control, and raising. We argue that copy control plays a crucial role in demarcating the division between functional and anaphoric control. Functional control relies on syntactic structure sharing to capture the interpretive dependency between controller and

²⁶In LFG, binding conditions are stated with reference to the f-structure rather than c-structure; thus, instead of c-command, LFG generally employs a notion of f-command in its binding theory (see, e.g., Dalrymple 1993; Lam 2021).

controllee, whereas anaphoric control treats the controllee as a pronominal element subject to binding constraints. Our formal analysis provides an explicit and precise account of the subtle empirical variations uncovered in our experiments. We draw insights from the recent advances in LFG’s theory of control, integrating elements of syntax, semantics, and discourse (PCDRT), while keeping the formal apparatus as lightweight as possible to enhance the accessibility of this work. Beyond its immediate empirical and analytical contributions, our analysis has broader theoretical implications. These include the contrast between grammatically licensed copy control and processing-based accounts of resumptions, the advantages of a lexicalist approach in the face of potential interspeaker variation, and how our analysis can contribute to modeling typological variation in copy control. To facilitate cross-framework dialogue, we have compared our analysis with the broader development of control theory in the Principles-and-Parameters and Minimalist traditions. We observe that the joint deployment of two control mechanisms in LFG—functional control, which resembles movement-based approaches, and anaphoric control, which parallels PRO-based approaches—within a single theoretical system to analyze complement control phenomena in a single language is distinctive in the broader cross-framework landscape of control theory. Taken together, these complementary mechanisms capture the rich diversity of control patterns observed both cross-linguistically and within individual languages. Mandarin copy control, as analyzed here, offers a clear illustration of how a hybrid model can provide a principled and empirically grounded account of complex control phenomena.

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